

5ME42A(ME)

Manufacturing

Technology

**Topic – Bulk Deformation Processes in
Metal Working**

Dr. Kailash Chaudhary

Ph.D. (Mechanical Design), M.E. (P & I), B.E. (Mechanical Engg)

Assistant Professor

Department of Mechanical Engineering

MBM University Jodhpur

Dr. Kailash Chaudhary, MBM University Jodhpur

Broad classification of Engineering Manufacturing Processes

1. Shaping or Forming

Manufacturing a solid product of definite size and shape from a given material taken in three possible states:

- **in liquid or semi-liquid state** – casting, injection moulding etc.
- **in solid state** – forging rolling, extrusion, drawing etc.
- **in powder form** – powder metallurgical process.

2. Joining Process

Welding, brazing, soldering etc.

3. Removal or Cutting Process

Machining (Traditional or Non-traditional),
Grinding etc.

4. Regenerative Manufacturing Process

Production of solid products in layer by layer from raw materials in different form:

- ❑ **liquid** – stereo lithography
- ❑ **powder** – selective sintering
- ❑ **sheet** – LOM (laminated object manufacturing)
- ❑ **wire** – FDM (Fused Deposition Modeling)

Regenerative manufacturing process is generally accomplished very rapidly and quite accurately using CAD and CAM for Rapid Prototyping and Tooling.

Metal Forming

Large group of manufacturing processes in which **plastic deformation** is used to change the shape of metal workpieces

The tool, usually called a **die**, applies stresses that **exceed the yield strength** of the metal

- The metal takes a shape determined by the geometry of the die

Bulk Deformation Processes In Metalworking

Rolling

Forging

Extrusion

Wire and Bar Drawing

Learning Outcomes

Learners will be able to:

1. Understand the basics of bulk deformation process
2. Differentiate among various bulk deformation processes
3. Understand features, operations and parameters of various processes

Bulk Deformation

Metal forming operations which cause *significant shape change* by deformation in metal parts whose *initial form is bulk rather than sheet* like cylindrical bars and rectangular slabs

- These processes work by stressing metal sufficiently to cause *plastic flow* into desired shape
- Performed as cold, warm, and hot working operations

Importance of Bulk Deformation

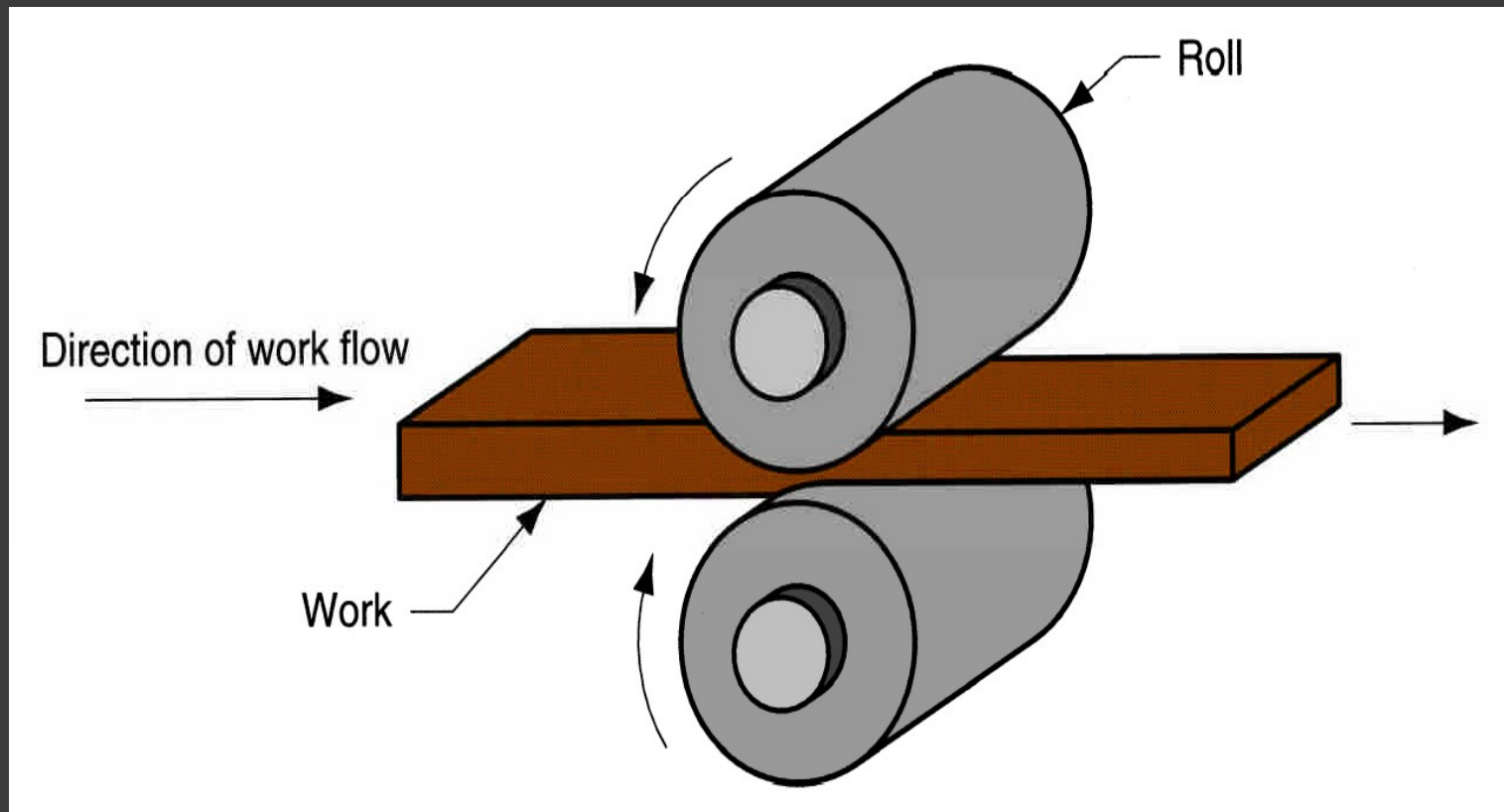
- In **hot working, significant shape change** can be accomplished ($>$ recrystallization temperature)
- In **cold working, strength can be increased** during shape change (at room temperature)
- **Little or no waste** - some operations are *near net shape* or *net shape* processes
 - The parts require little or no subsequent machining

Basic Bulk Deformation Processes

1. **Rolling** – slab or plate is squeezed **between opposing rolls**
2. **Forging** – work is squeezed and shaped **between opposing dies**
3. **Extrusion** – work is squeezed through a **die opening**, thereby taking the shape of the opening
4. **Wire and bar drawing** – diameter of wire or bar is reduced by **pulling it through a die opening**

Rolling

work thickness is reduced by compressive forces exerted by two opposing rolls



Types of Rolling

- By geometry of work:
 - *Flat rolling* - used to reduce thickness of a rectangular cross-section
 - *Shape rolling* - a square cross-section is formed into a shape such as an I-beam
- By temperature of work:
 - *Hot Rolling*
 - *Cold rolling*

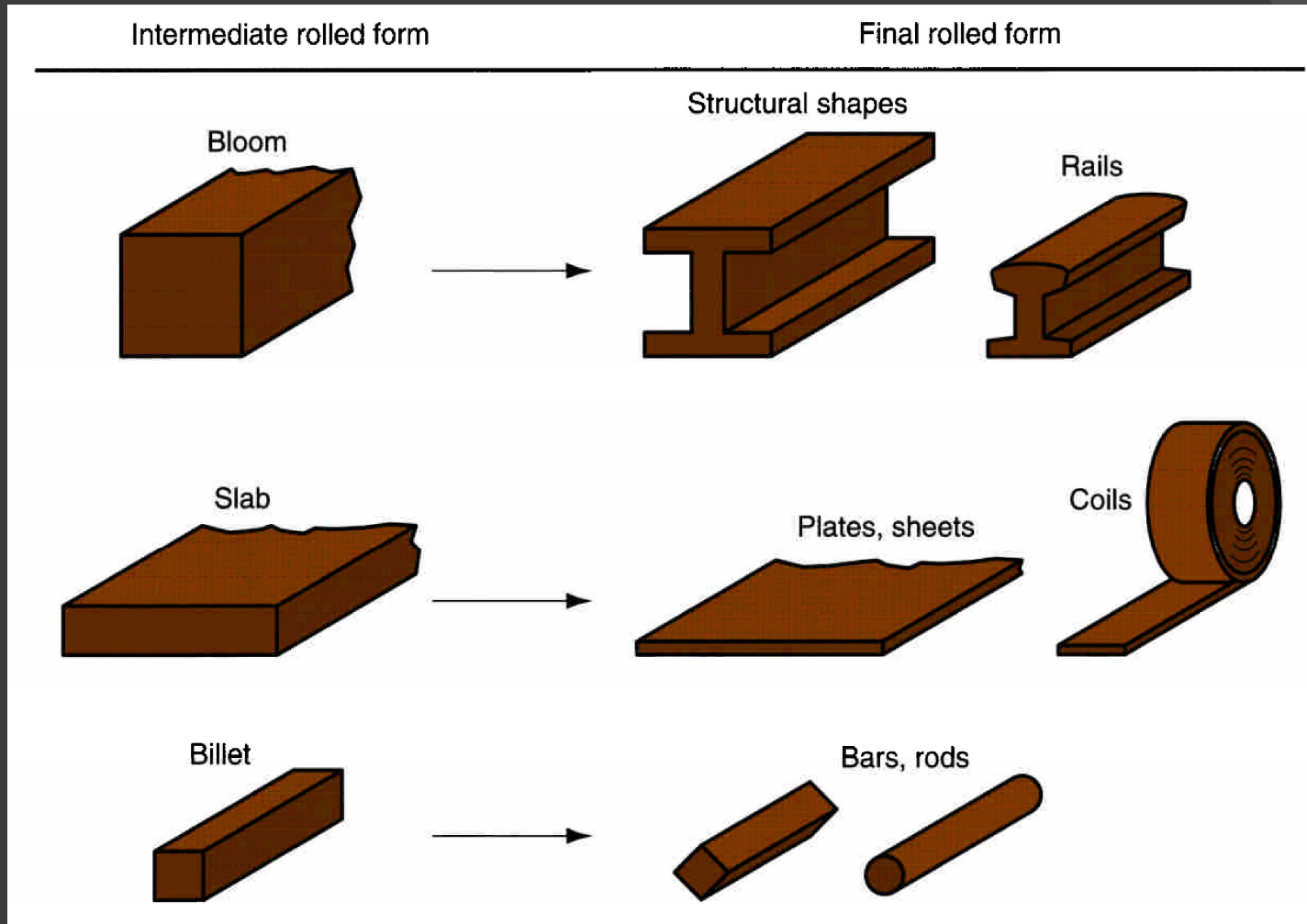


Figure - Some of the steel products made in a rolling mill

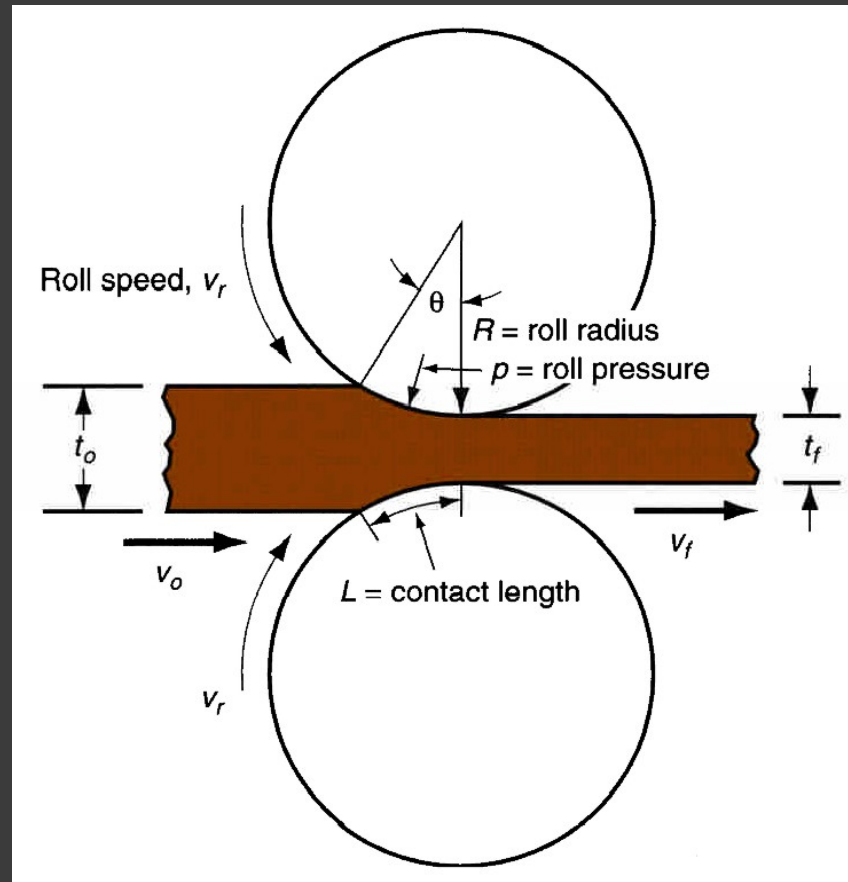


Figure - Side view of flat rolling, indicating before and after thicknesses, work velocities, angle of contact with rolls, and other features

Flat Rolling – Terminology

Draft = amount of thickness reduction

$$d = t_o - t_f$$

where d = draft; t_o = starting thickness; and t_f = final thickness

Reduction = draft expressed as a fraction of starting stock thickness:

$$r = \frac{d}{t_o}$$

where r = reduction

Shape Rolling

Work is deformed into a contoured cross-section rather than flat (rectangular)

- Accomplished by passing work through rolls that have the reverse of desired shape
- Products include:
 - Construction shapes such as I-beams, L-beams, and U-channels
 - Rails for railroad tracks
 - Round and square bars and rods



Picture of a rolling mill for hot flat rolling; the steel plate is seen as the glowing strip extending diagonally from the lower left corner

(photo courtesy of Bethlehem Steel Company)

Rolling Mills

- Equipment is massive and expensive
- Rolling mill configurations:
 - Two-high – two opposing large diameter rolls
 - Three-high – work passes through both directions
 - Four-high – backing rolls support smaller work rolls
 - Cluster mill – multiple backing rolls on smaller rolls
 - Tandem rolling mill – sequence of two-high mills

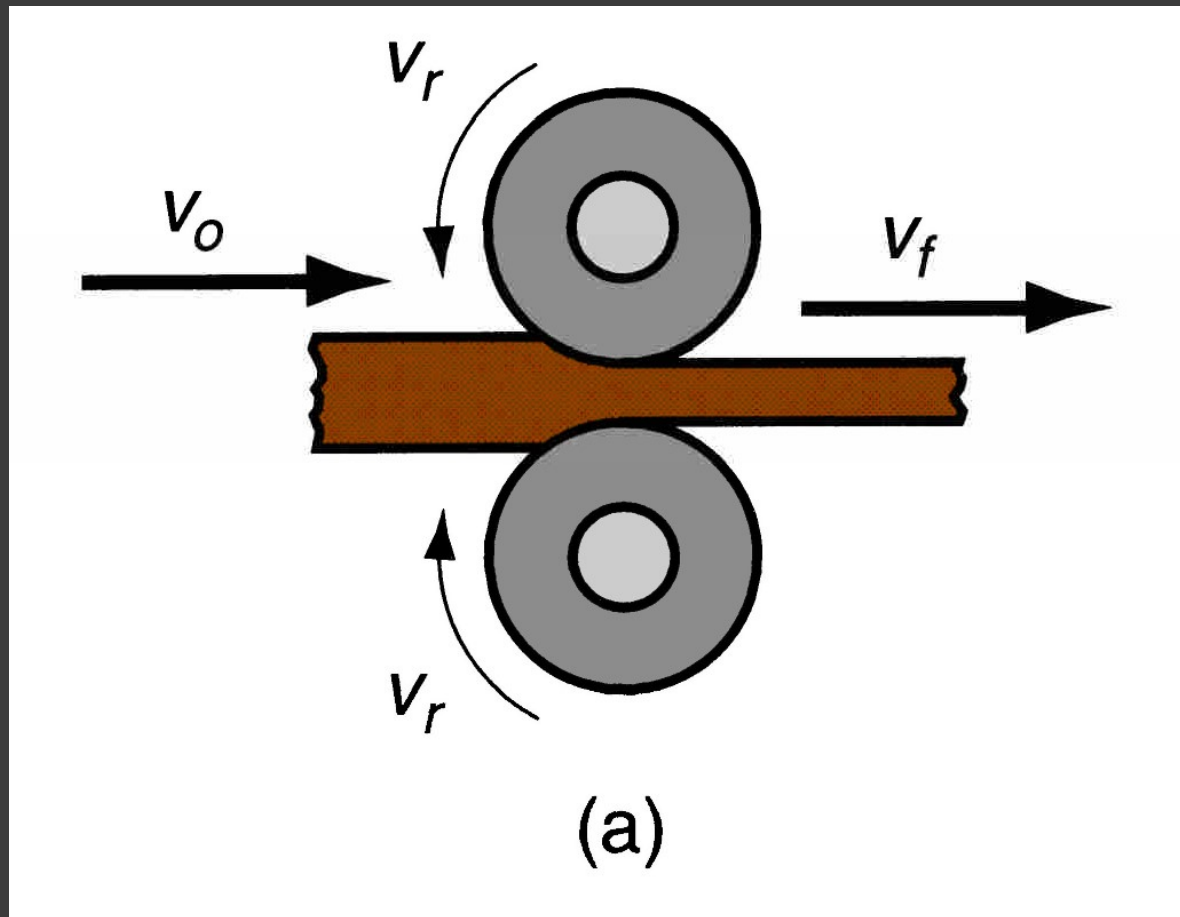


Figure - (a) 2-high rolling mill

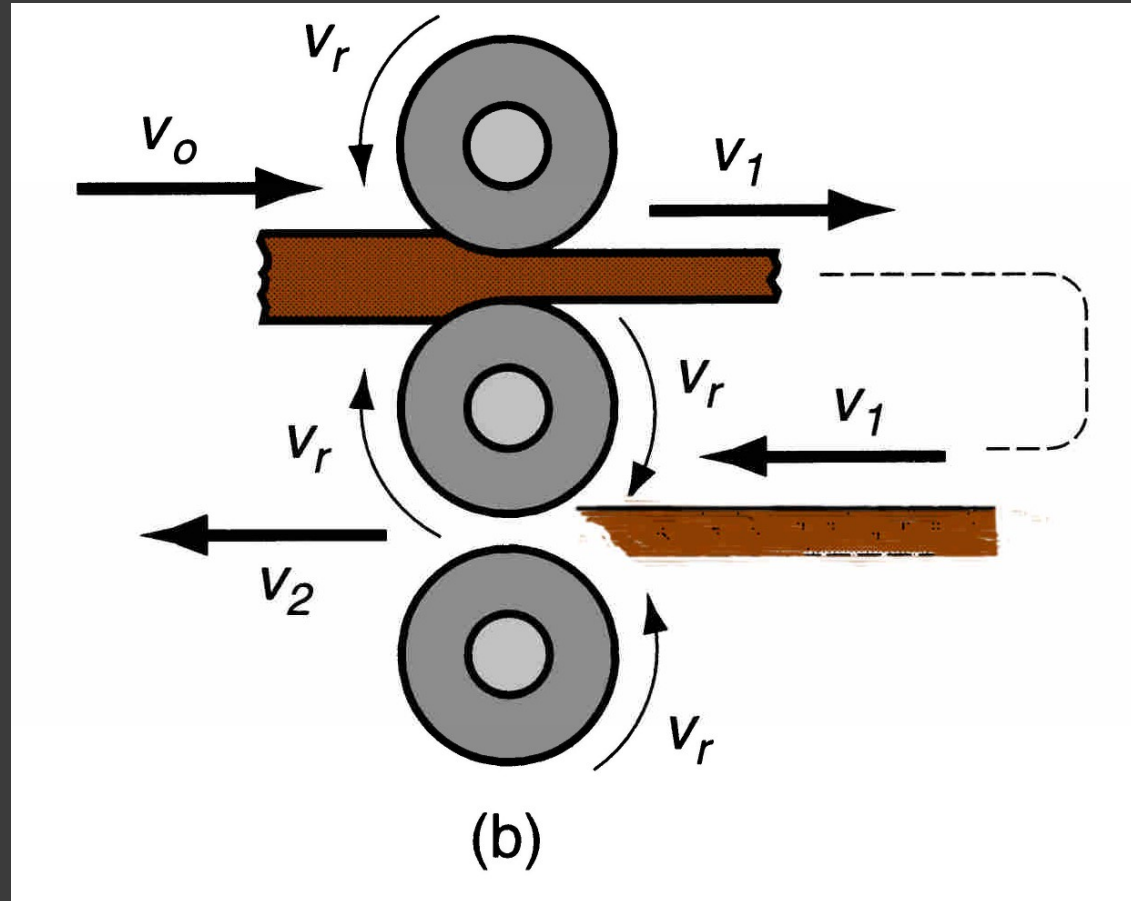


Figure - (b) 3-high rolling mill

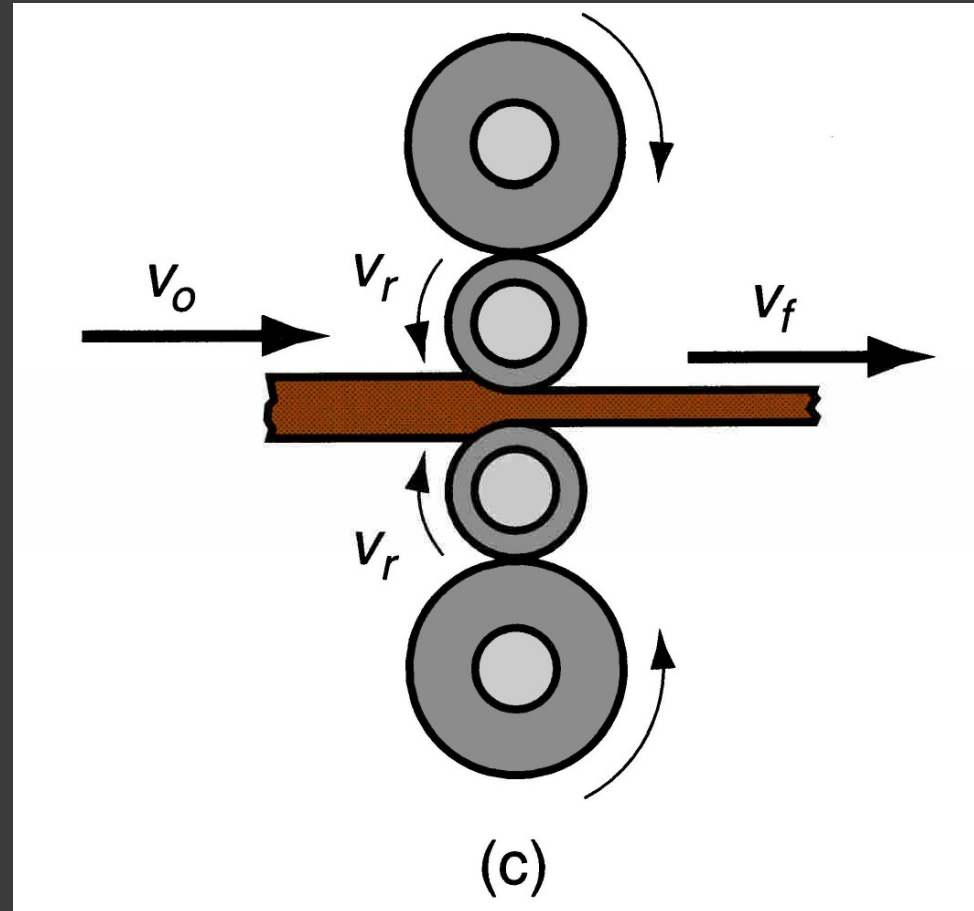


Figure - (c) four-high rolling mill

Cluster Mill

Multiple backing rolls allow even smaller roll diameters

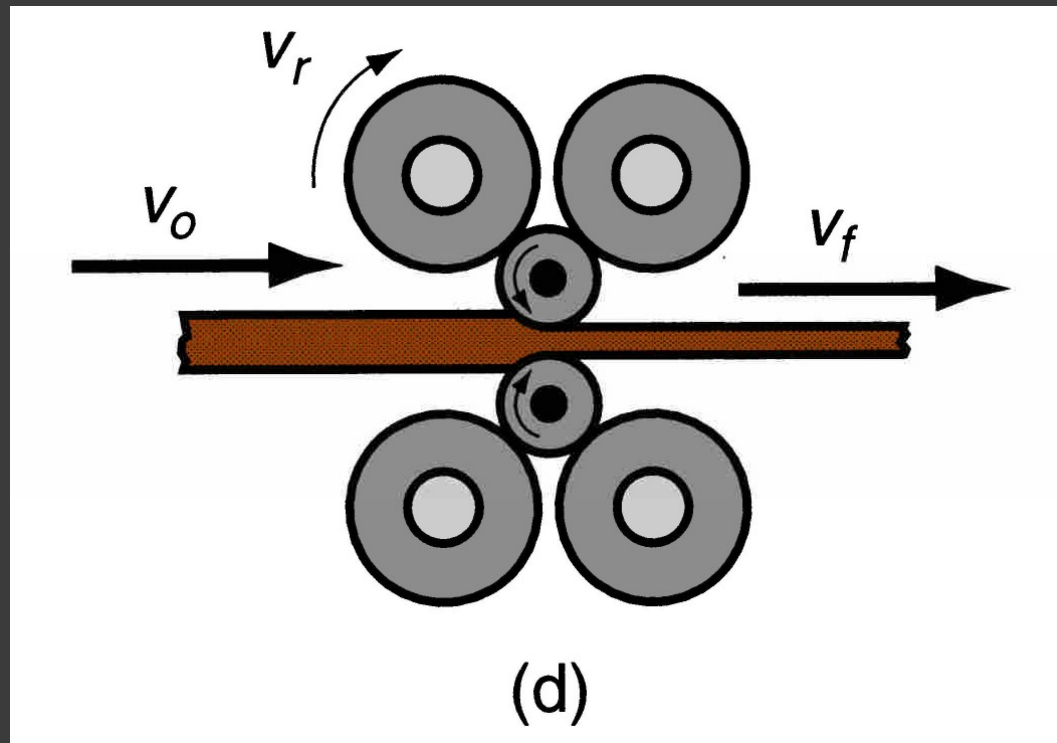


Figure - (d) cluster mill

Tandem Rolling Mill

A series of rolling stands in sequence

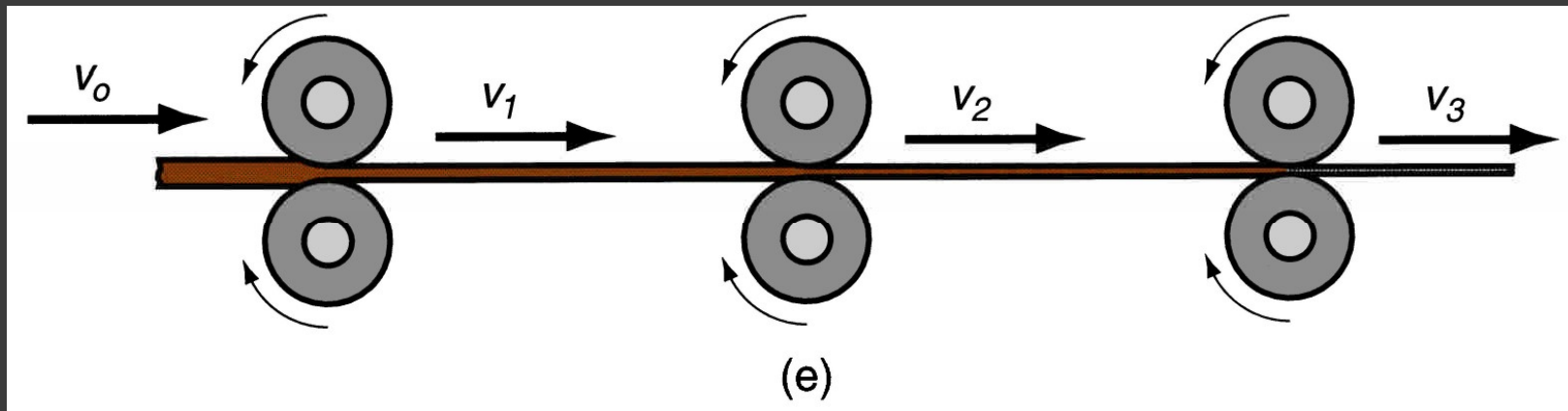


Figure - (e) tandem rolling mill

Thread Rolling

Bulk deformation process used to form threads on cylindrical parts by rolling them between two dies

- Most important commercial process for mass producing bolts and screws
- Performed by cold working in thread rolling machines
- Advantages over thread cutting (machining):
 - Higher production rates
 - Better material utilization
 - Stronger threads due to work hardening
 - Better fatigue resistance due to compressive stresses introduced by rolling

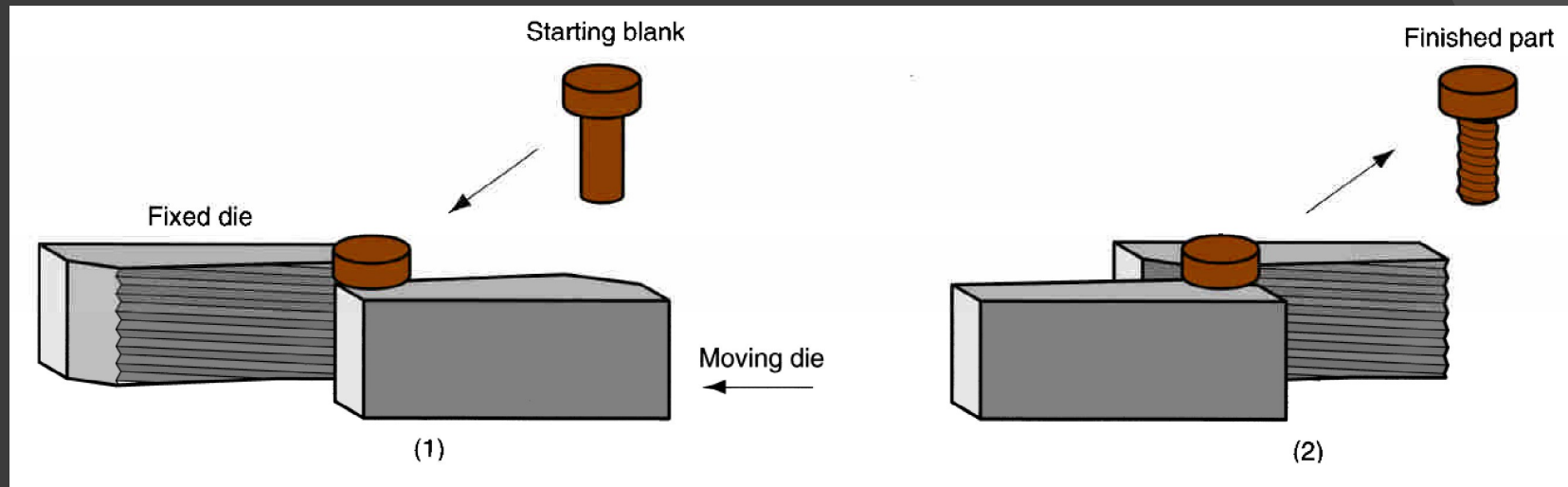


Figure - Thread rolling with flat dies:
(1) start of cycle, and (2) end of cycle

Ring Rolling

Deformation process in which a thick-walled ring of smaller diameter is rolled into a thin-walled ring of larger diameter

- As thick-walled ring is compressed, deformed metal elongates, causing diameter of ring to be enlarged
- Hot working process for large rings and cold working process for smaller rings
- Applications: ball and roller bearing races, steel tires for railroad wheels, and rings for pipes, pressure vessels, and rotating machinery
- Advantages: material savings, ideal grain orientation, strengthening through cold working

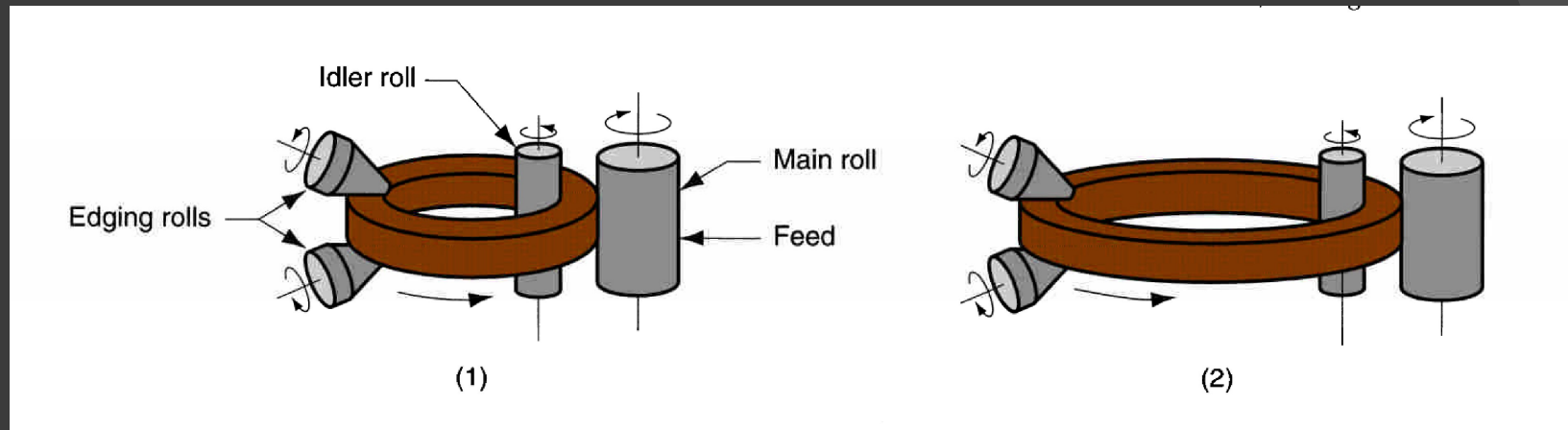


Figure - Ring rolling used to reduce the wall thickness and increase the diameter of a ring:

(1) start, and (2) completion of process

Forging

Deformation process in which work is compressed between two dies

- Oldest of the metal forming operations, dating from about 5000 B C
- Components: engine crankshafts, connecting rods, gears, aircraft structural components, jet engine turbine parts
- In addition, basic metals industries use forging to establish basic form of large components that are subsequently machined to final shape and size

Classification of Forging Operations

- Cold vs. hot forging:
 - *Hot or warm forging* – most common, due to the significant deformation and the need to reduce strength and increase ductility of work metal
 - *Cold forging* - advantage is increased strength that results from strain hardening
- Impact vs. press forging:
 - *Forge hammer* - applies an impact load
 - *Forge press* - applies gradual pressure

Types of Forging Dies

- *Open-die forging* - work is compressed between two flat dies, allowing metal to flow laterally without constraint
- *Impression-die forging* - die surfaces contain a cavity or impression that is imparted to workpart, thus constraining metal flow - *flash* is created
- *Flashless forging* - workpart is completely constrained in die and no excess flash is produced

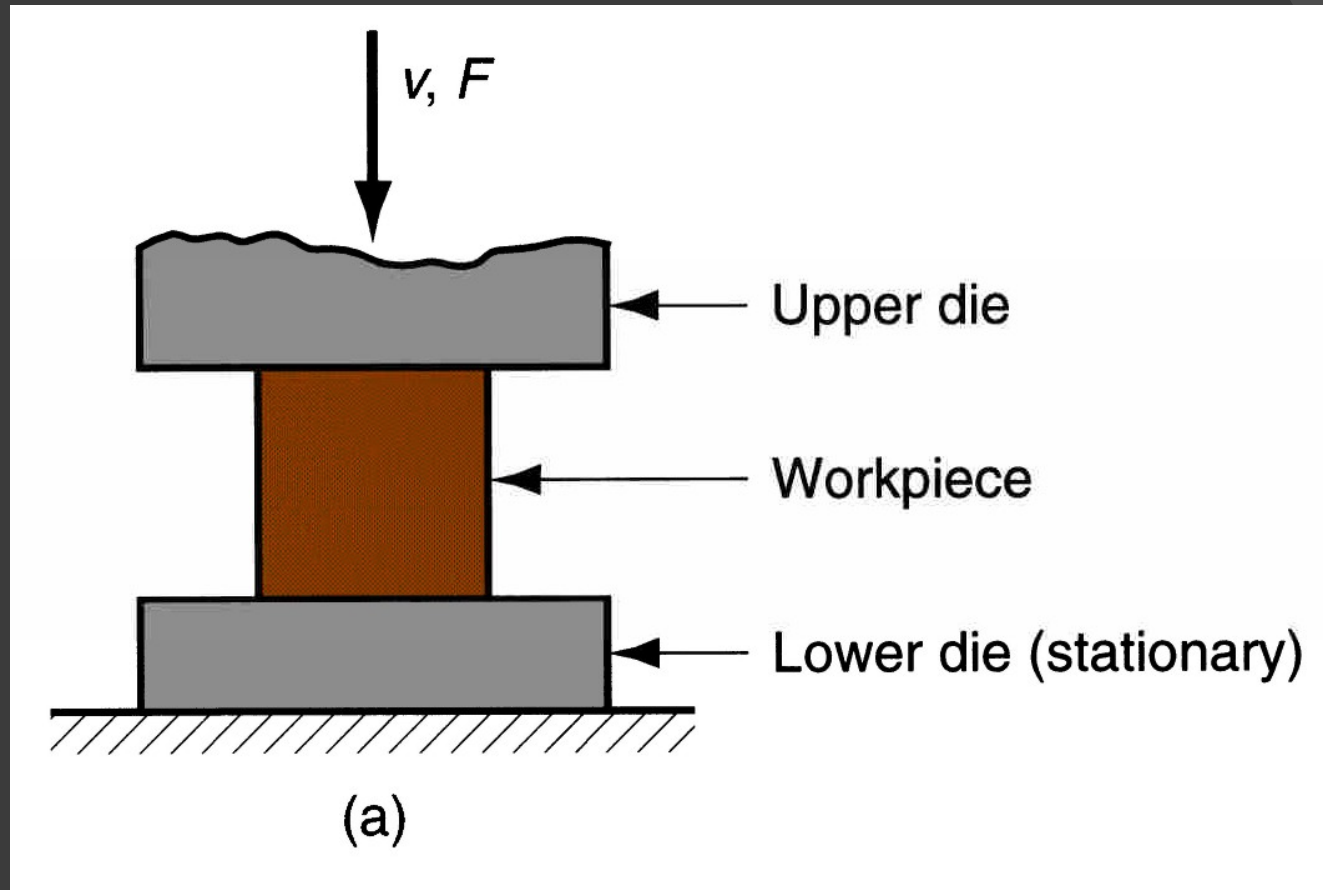


Figure - Three types of forging: (a) open-die forging

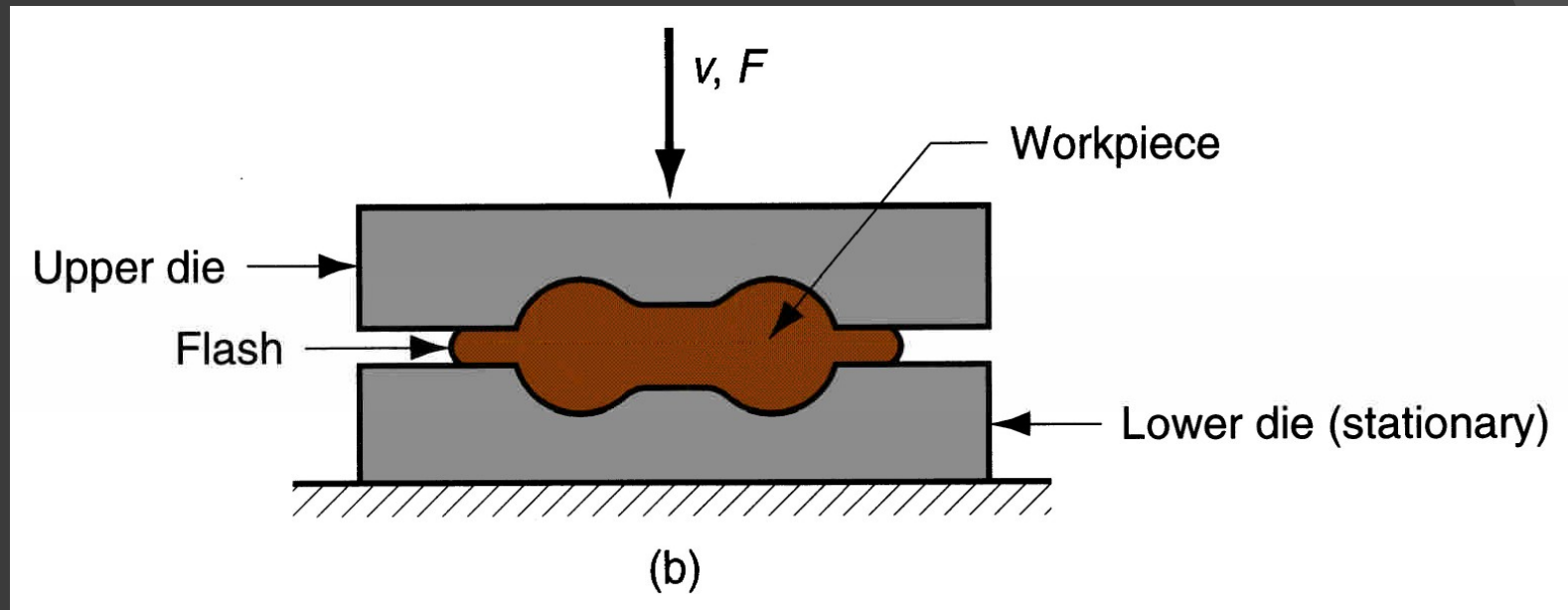


Figure - Three types of forging (b) impression-die forging

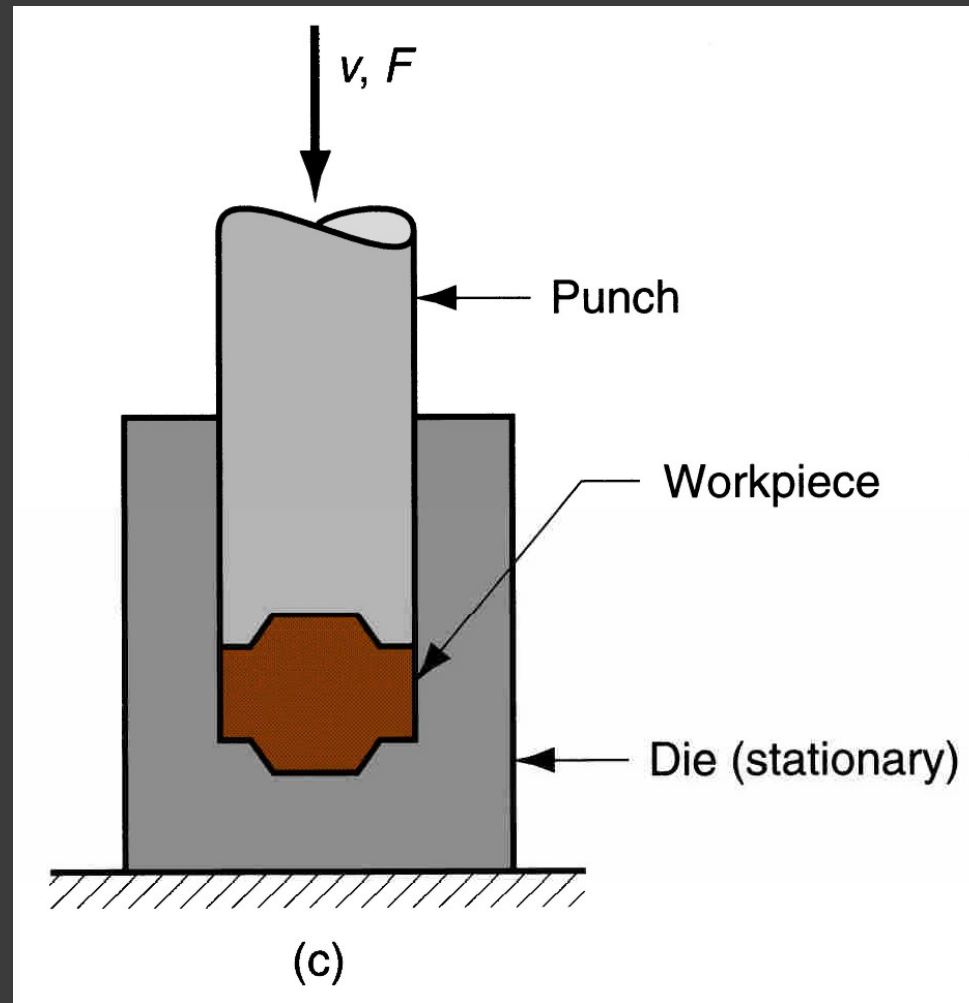


Figure - Three types of forging (c) flashless forging

Open-Die Forging

Compression of workpart with cylindrical cross-section between two flat dies

- Similar to compression test
- Deformation operation reduces height and increases diameter of work
- Common names include *upsetting* or *upset forging*

Open-Die Forging with No Friction

If no friction occurs between work and die surfaces, then homogeneous deformation occurs, so that radial flow is uniform throughout workpart height and true strain is given by:

$$\varepsilon = \ln \frac{h_o}{h}$$

where h_o = starting height; and h = height at some point during compression

- At h = final value h_f , true strain is maximum value

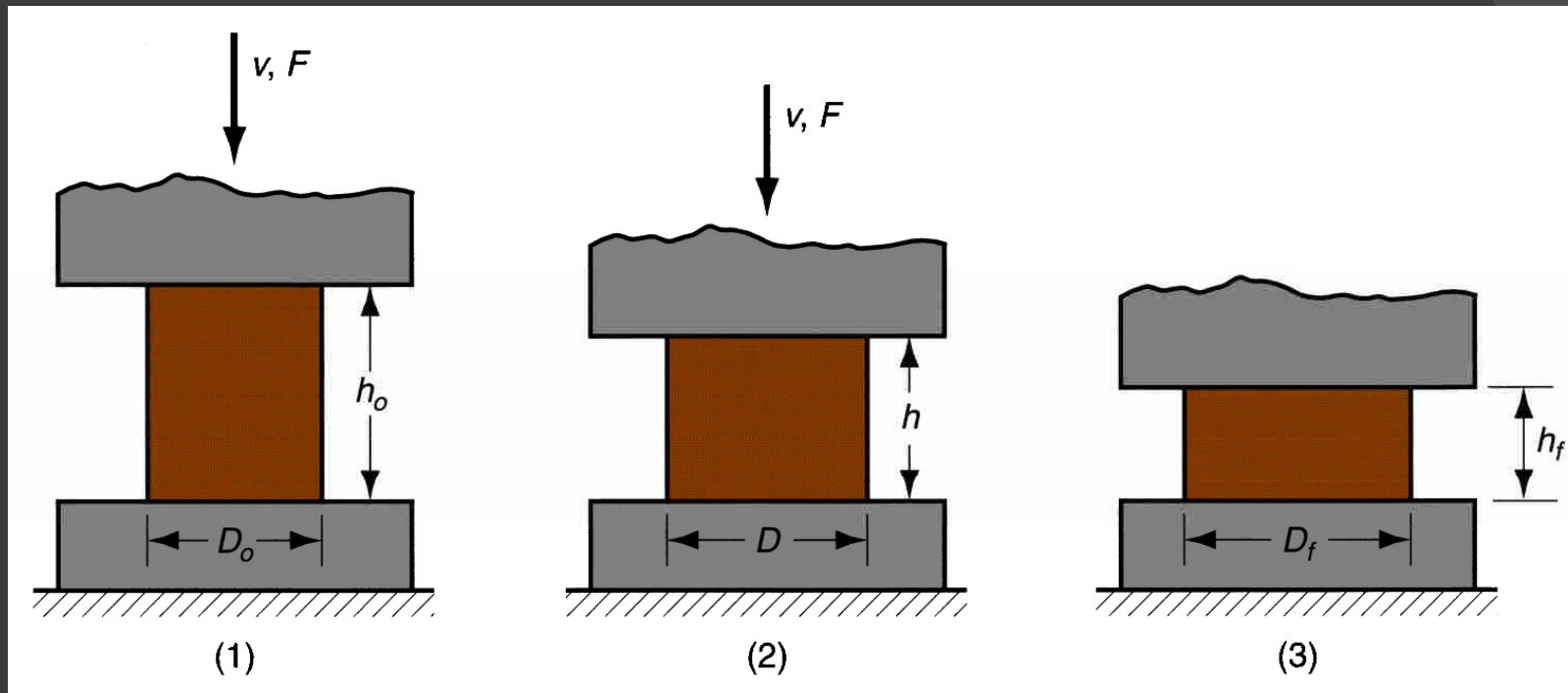


Figure - Homogeneous deformation of a cylindrical workpart under ideal conditions in an open-die forging operation:
(1) start of process with workpiece at its original length and diameter, (2) partial compression, and (3) final size

Open-Die Forging with Friction

- Friction between work and die surfaces constrains lateral flow of work, resulting in barreling effect
- In hot open-die forging, effect is even more pronounced due to heat transfer at and near die surfaces, which cools the metal and increases its resistance to deformation

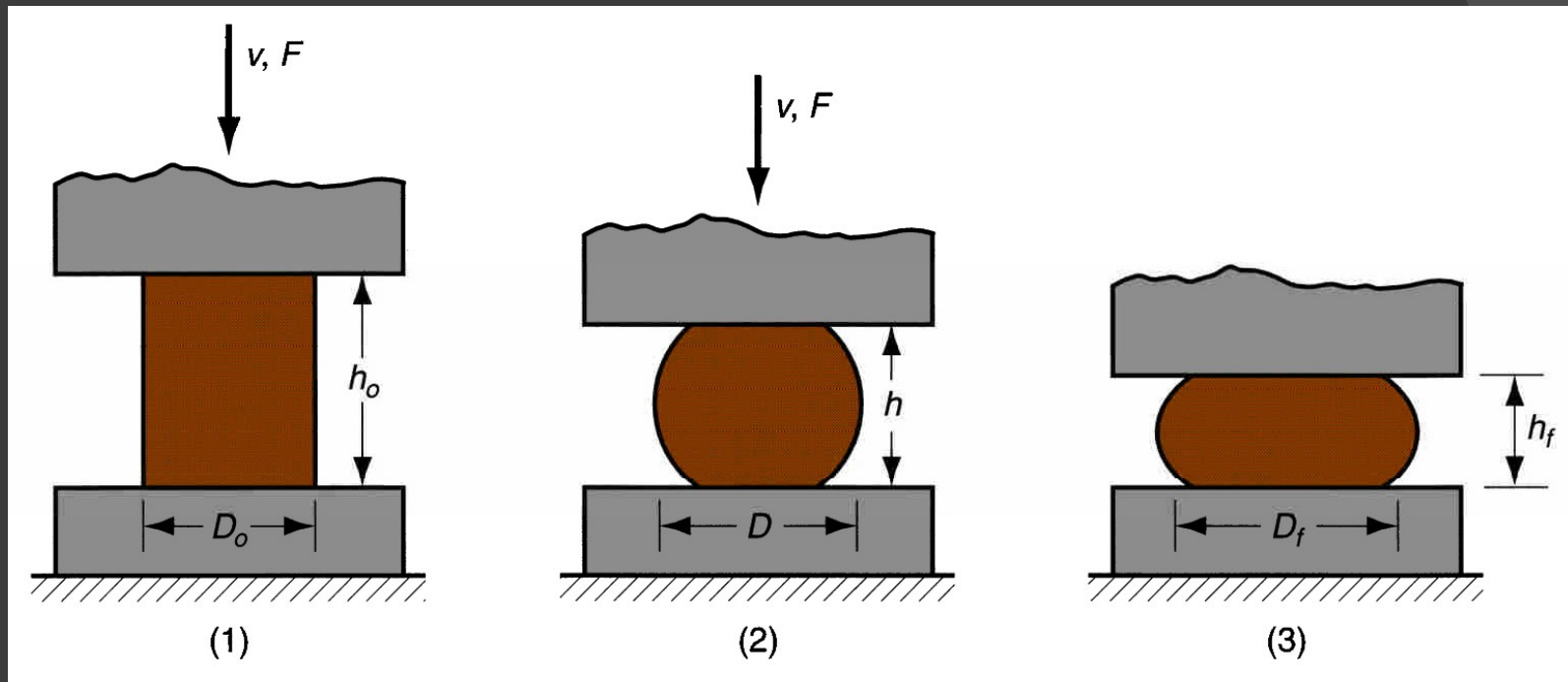


Figure - Actual deformation of a cylindrical workpart in open-die forging, showing pronounced *barreling*:
(1) start of process, (2) partial deformation, and (3) final shape

Impression-Die Forging

Compression of workpart by dies with inverse of desired part shape

- Flash is formed by metal that flows beyond die cavity into small gap between die plates
- Flash must be later trimmed from part, but it serves an important function during compression:
 - As flash forms, friction resists continued metal flow into gap, constraining material to fill die cavity
 - In hot forging, metal flow is further restricted by cooling against die plates

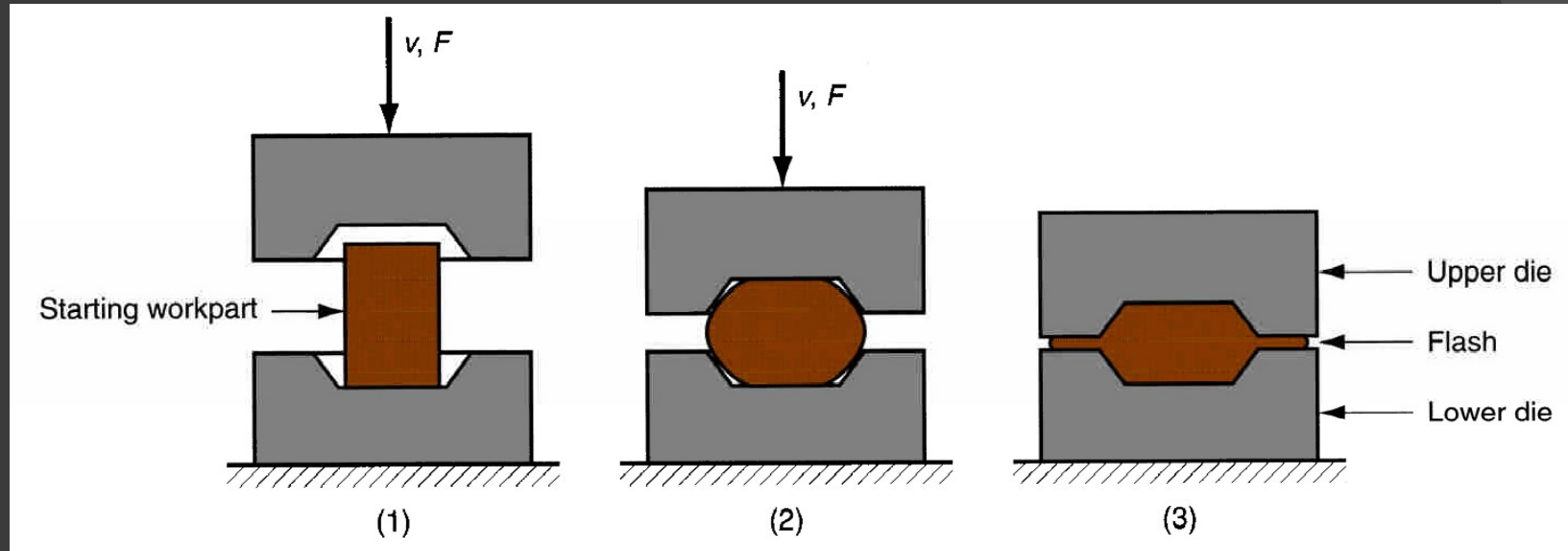


Figure - Sequence in impression-die forging:

- (1) just prior to initial contact with raw workpiece,
- (2) partial compression, and
- (3) final die closure, causing flash to form in gap between die plates

Impression-Die Forging Practice

- Several forming steps often required, with separate die cavities for each step
 - Beginning steps redistribute metal for more uniform deformation and desired metallurgical structure in subsequent steps
 - Final steps bring the part to its final geometry
 - Impression-die forging is often performed manually by skilled operator under adverse conditions

Impression-Die Forging

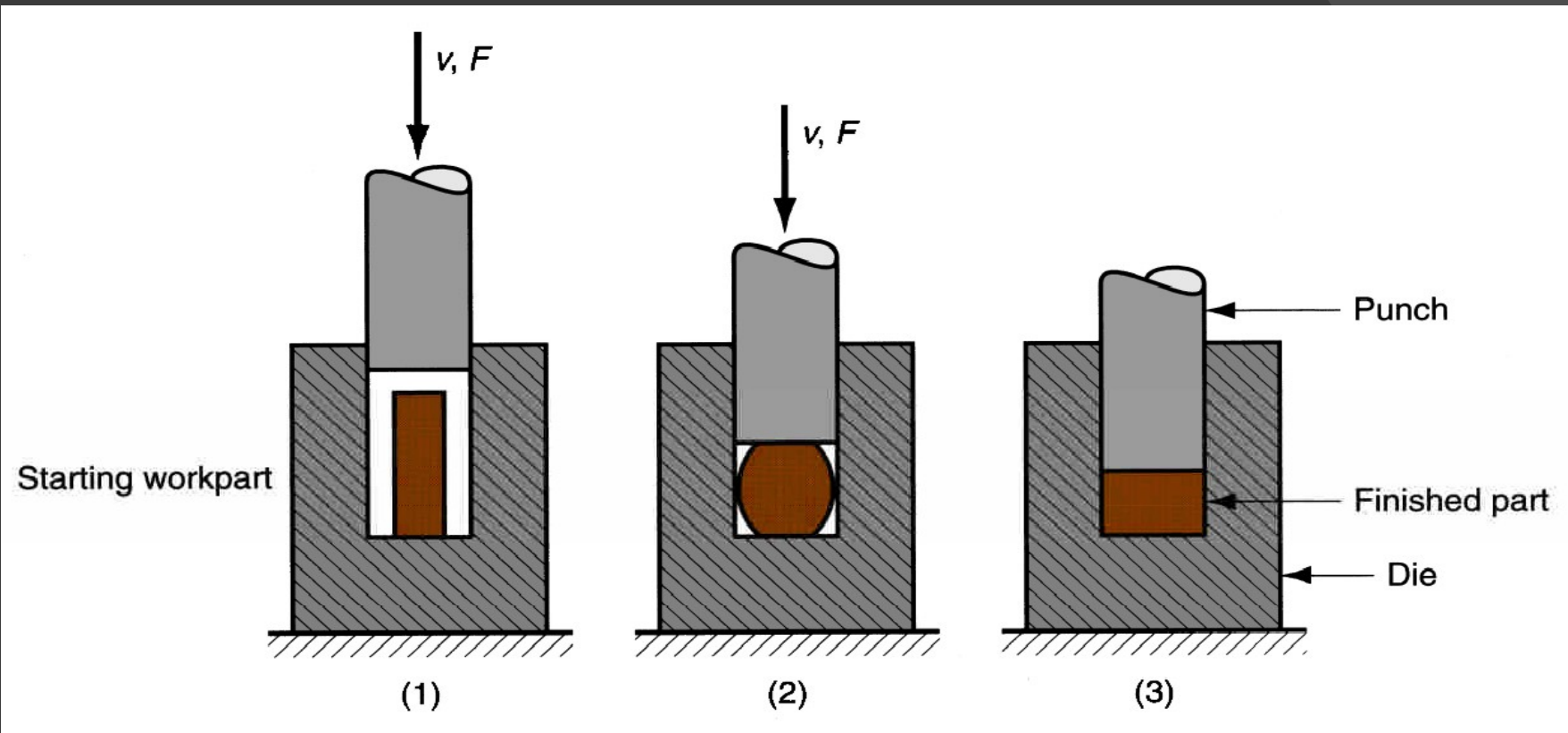
Advantages and Limitations

- Advantages compared to machining from solid stock:
 - Higher production rates
 - Conservation of metal (less waste)
 - Greater strength
 - Favorable grain orientation in the metal
- Limitations:
 - Not capable of close tolerances
 - Machining often required to achieve accuracies and features needed, such as holes, threads, and mating surfaces that fit with other components

Flashless Forging

Compression of work in punch and die tooling whose cavity does allow for flash

- Starting workpart volume must equal die cavity volume within very close tolerance
- Process control more demanding than impression-die forging
- Best suited to part geometries that are simple and symmetrical
- Often classified as a *precision forging* process

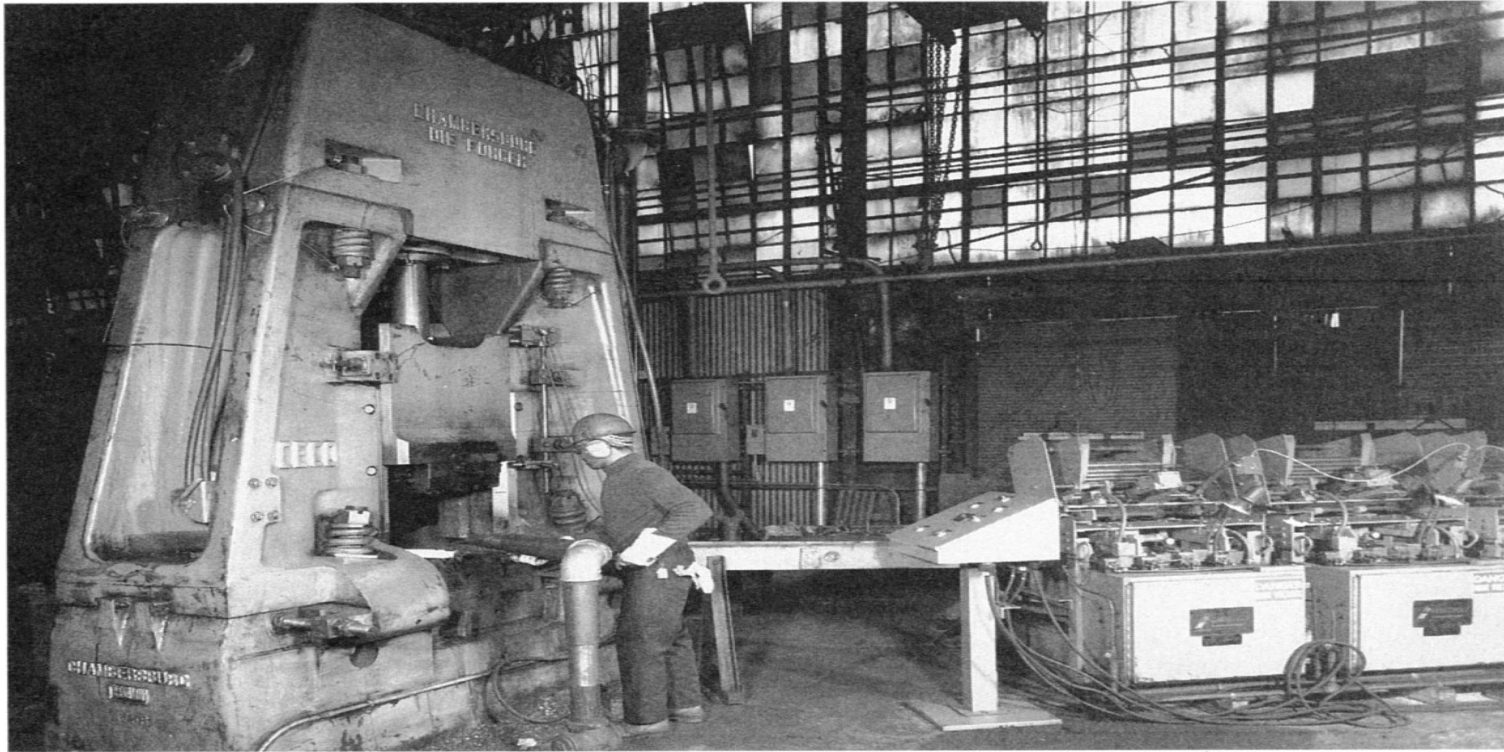


Flashless forging

- (1) just before initial contact with workpiece,
- (2) partial compression, and
- (3) final punch and die closure

Forging Hammers (Drop Hammers)

- Apply an impact load against workpart - two types:
 - *Gravity drop hammers* - impact energy from falling weight of a heavy ram
 - *Power drop hammers* - accelerate the ram by pressurized air or steam
- Disadvantage: impact energy transmitted through anvil into floor of building
- Most commonly used for impression-die forging



Drop forging hammer, fed by conveyor and heating units at the right of the scene (photo courtesy of Chambersburg Engineering Company)

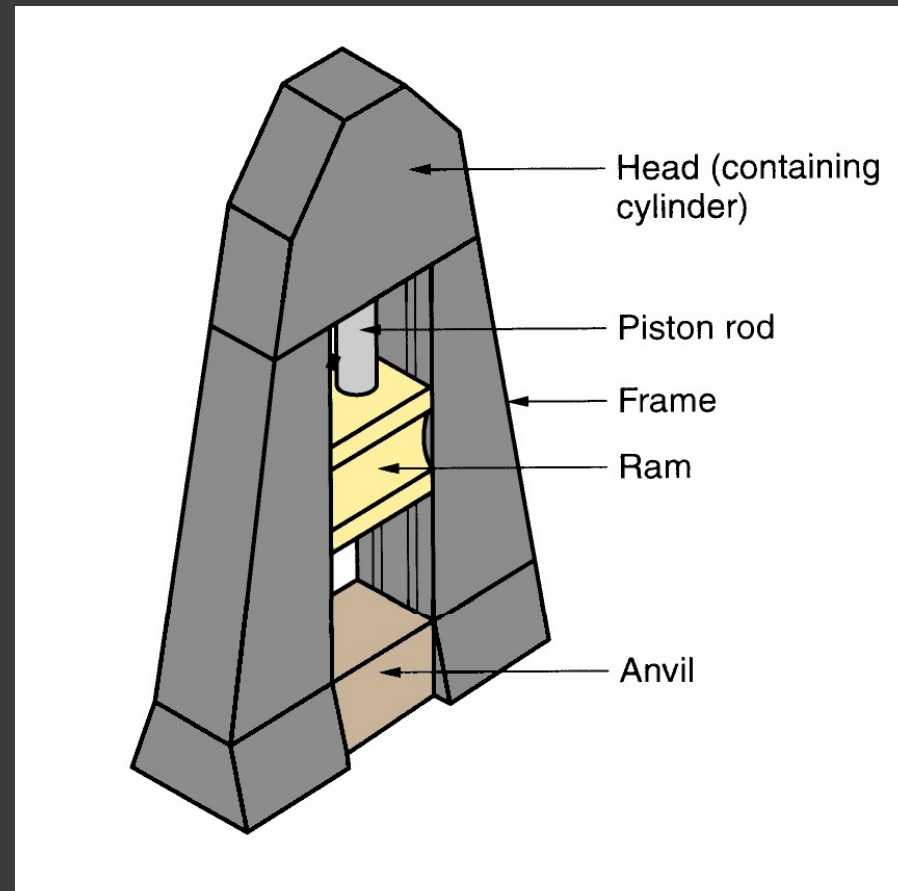


Diagram showing details of a drop hammer for impression-die forging

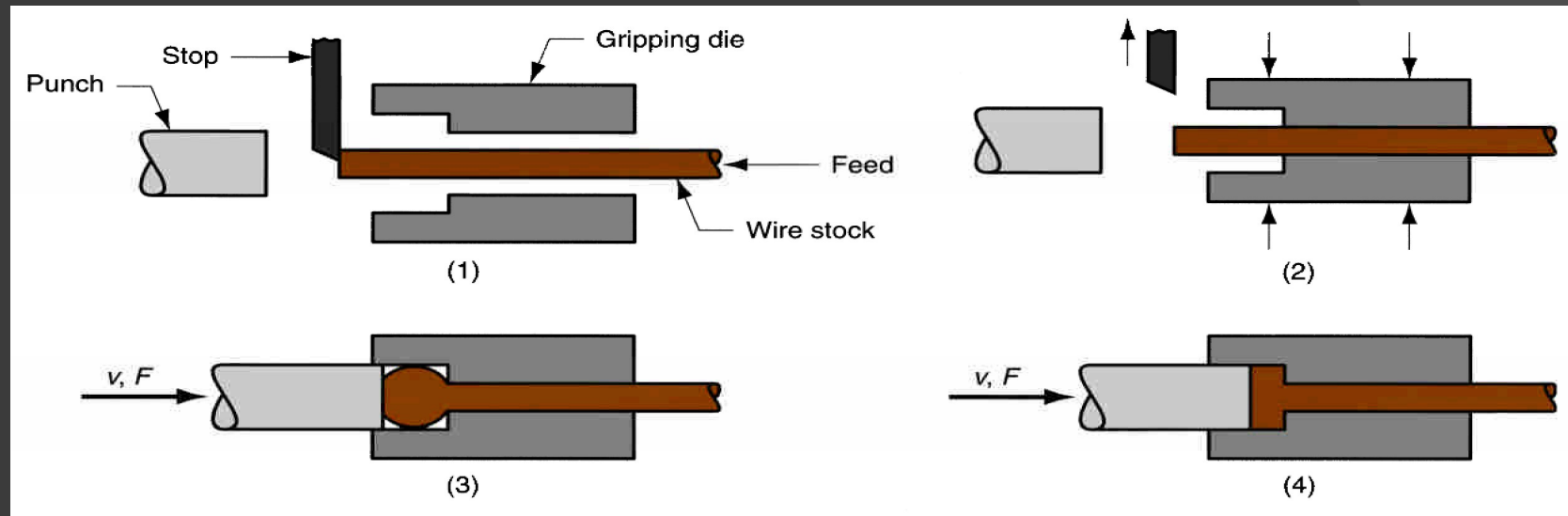
Forging Presses

- Apply gradual pressure to accomplish compression operation - types:
 - *Mechanical presses* - converts rotation of drive motor into linear motion of ram
 - *Hydraulic presses* - hydraulic piston actuates ram
 - *Screw presses* - screw mechanism drives ram

Upsetting and Heading

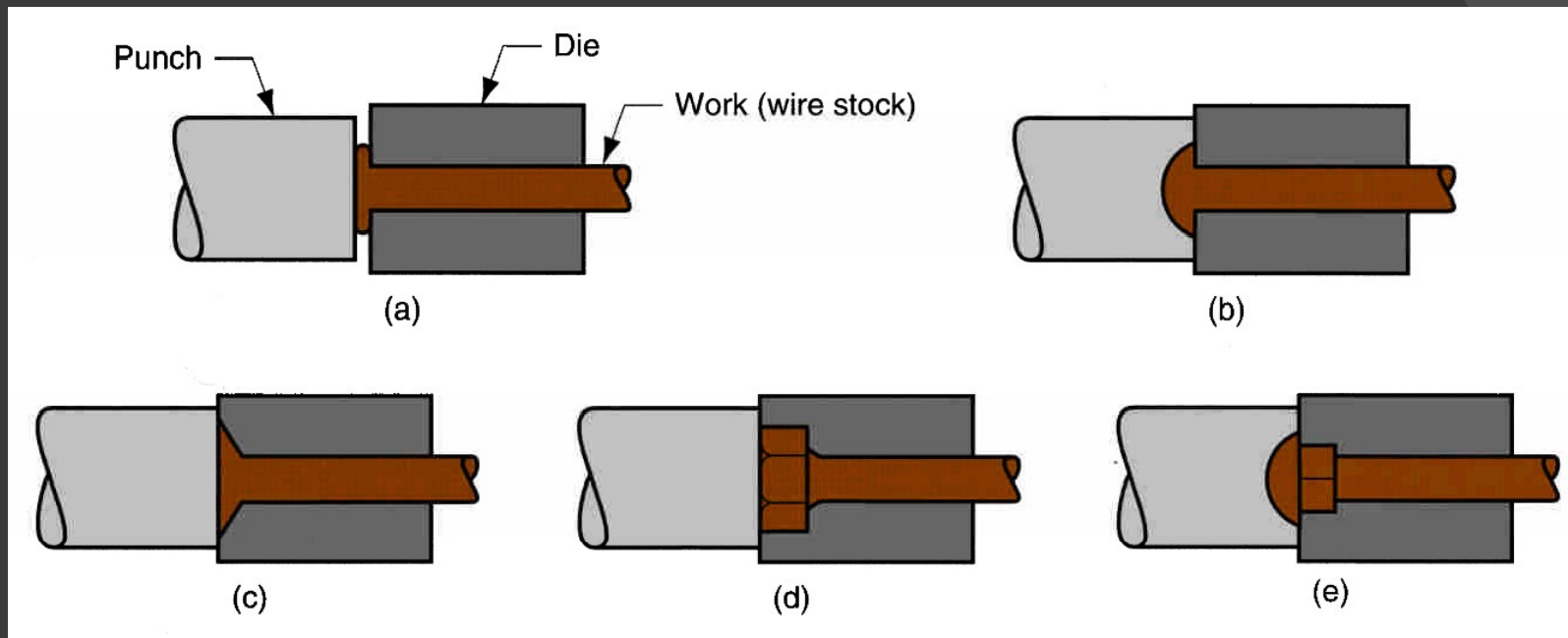
Forging process used to form heads on nails, bolts, and similar hardware products

- More parts produced by upsetting than any other forging operation
- Performed cold, warm, or hot on machines called *headers* or *formers*
- Wire or bar stock is fed into machine, end is headed, then piece is cut to length
- For bolts and screws, thread rolling is then used to form threads



An upset forging operation to form a head on a bolt or similar hardware item. The cycle consists of:

- (1) wire stock is fed to the stop
- (2) gripping dies close on the stock and the stop is retracted
- (3) punch moves forward
- (4) bottoms to form the head



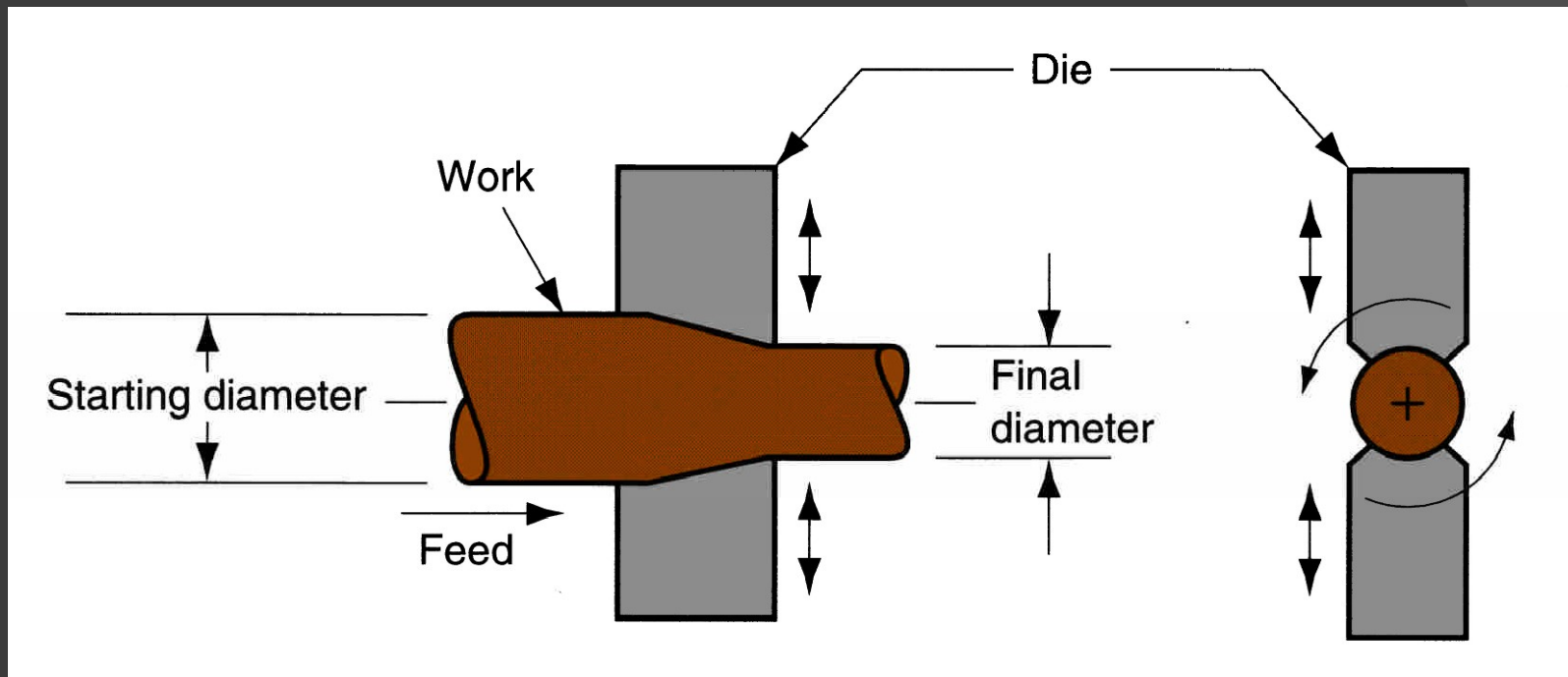
Examples of heading (upset forging) operations:

- (a) heading a nail using open dies
- (b) round head formed by punch
- (c) and (d) two common head styles for screws formed by die
- (e) carriage bolt head formed by punch and die

Swaging

Accomplished by rotating dies that hammer a workpiece radially inward to taper it as the piece is fed into the dies

- Used to reduce diameter of tube or solid rod stock
- Mandrel sometimes required to control shape and size of internal diameter of tubular parts



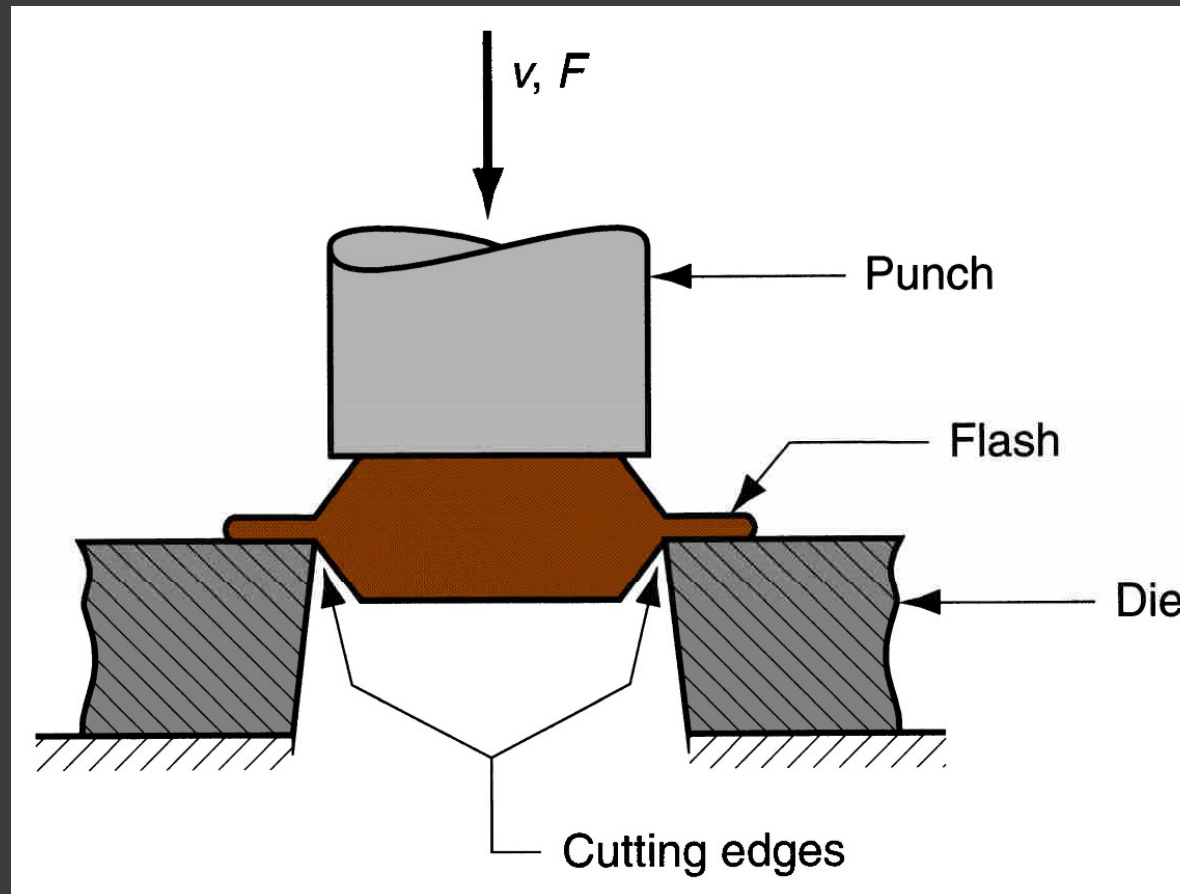
Swaging process to reduce solid rod stock; the dies rotate as they hammer the work

In radial forging, the workpiece rotates while the dies remain in a fixed orientation as they hammer the work

Trimming

Cutting operation to remove flash from workpart in impression-die forging

- Usually done while work is still hot, so a separate trimming press is included at the forging station
- Trimming can also be done by alternative methods, such as grinding or sawing



Trimming operation (shearing process) to remove the flash after impression-die forging

Extrusion

Compression forming process in which the work metal is forced to flow through a die opening to produce a desired cross-sectional shape

- Process is similar to squeezing toothpaste out of a toothpaste tube
- In general, extrusion is used to produce long parts of uniform cross-sections
- Two basic types of extrusion:
 - Direct extrusion
 - Indirect extrusion

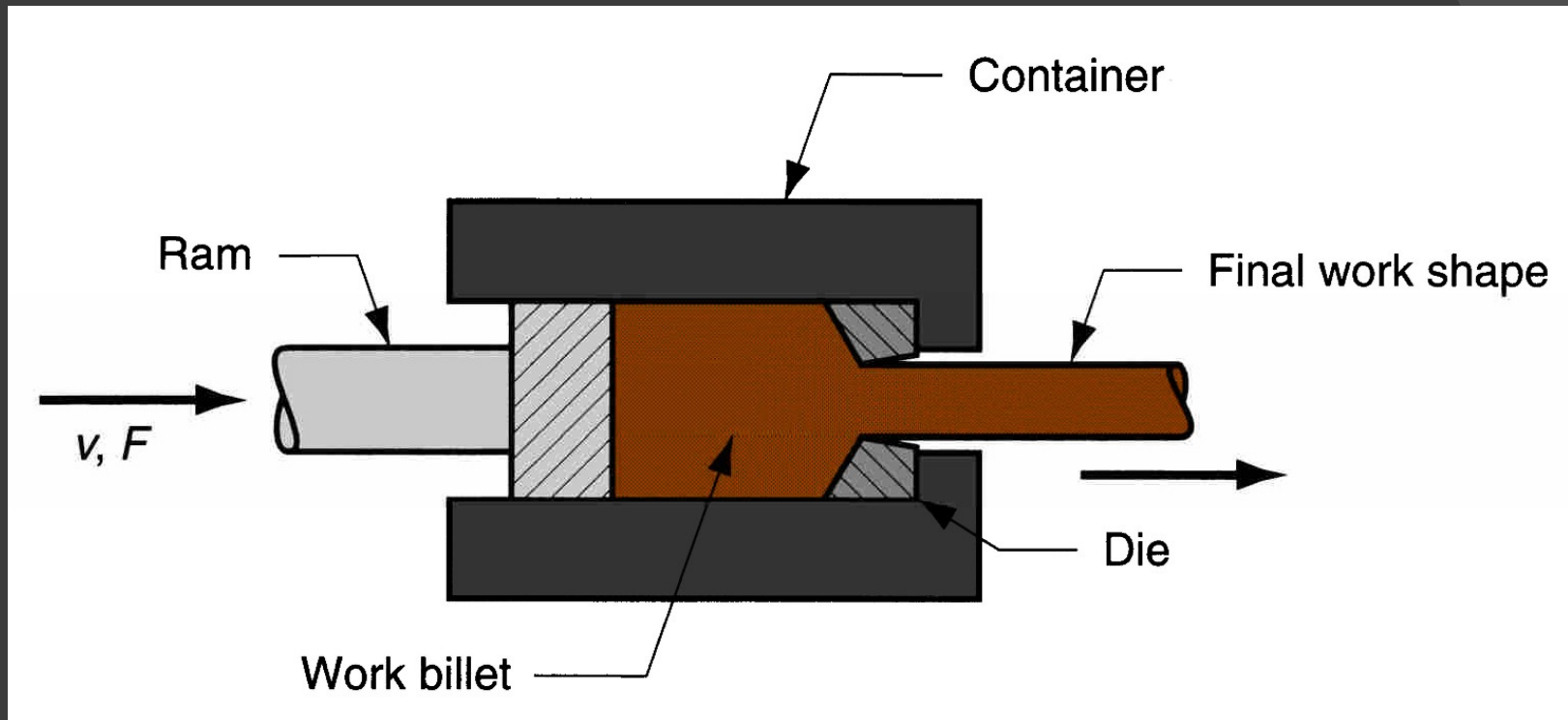


Figure - Direct extrusion

Comments on Direct Extrusion

- Also called *forward extrusion*
- As ram approaches die opening, a small portion of billet remains that cannot be forced through die opening
- This extra portion, called the *butt*, must be separated from extruded product by cutting it just beyond the die exit
- Starting billet cross section usually round, but final shape is determined by die opening

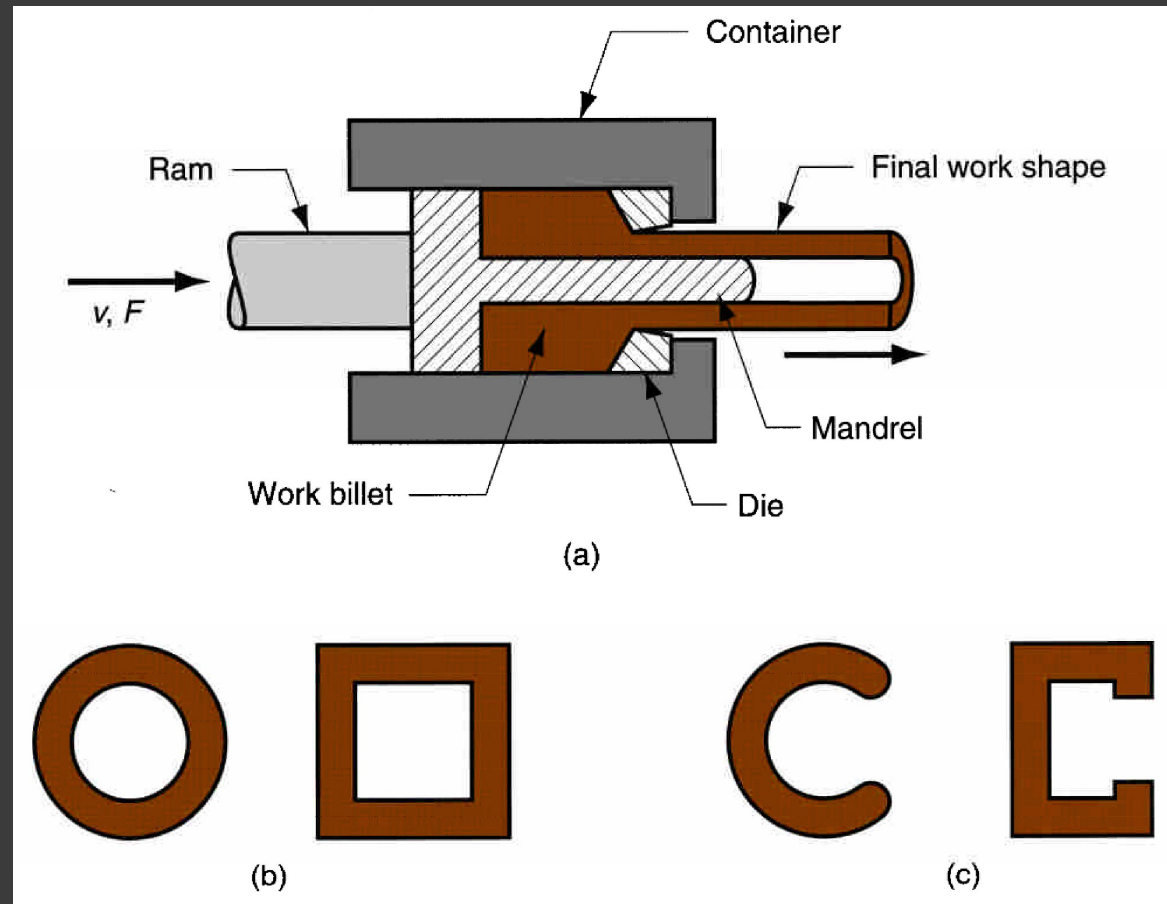


Figure - (a) Direct extrusion to produce a hollow or semi-hollow cross-section; (b) hollow and (c) semi-hollow cross- sections

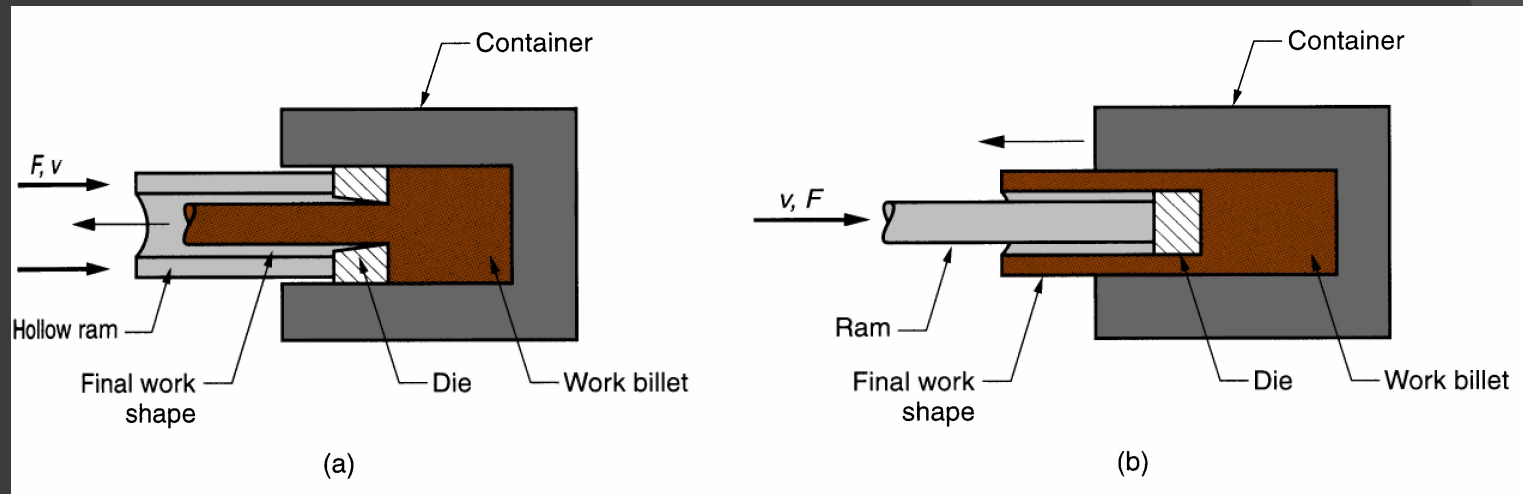


Figure - Indirect extrusion to produce
(a) a solid cross-section and (b) a hollow cross-section

Comments on Indirect Extrusion

- Also called *backward extrusion* and *reverse extrusion*
- Limitations of indirect extrusion are imposed by the lower rigidity of hollow ram and difficulty in supporting extruded product as it exits die

General Advantages of Extrusion

- Variety of shapes possible, especially in hot extrusion
 - Limitation: part cross-section must be uniform throughout length
- Grain structure and strength enhanced in cold and warm extrusion
- Close tolerances possible, especially in cold extrusion
- In some operations, little or no waste of material

Hot vs. Cold Extrusion

- *Hot extrusion* - prior heating of billet to above its recrystallization temperature
 - This reduces strength and increases ductility of the metal, permitting more size reductions and more complex shapes
- *Cold extrusion* - generally used to produce discrete parts
 - The term *impact extrusion* is used to indicate high speed cold extrusion

Extrusion Ratio

Also called the *reduction ratio*, it is defined as

$$r_x = \frac{A_o}{A_f}$$

where r_x = extrusion ratio; A_o = cross-sectional area of the starting billet; and A_f = final cross-sectional area of the extruded section

Applies to both direct and indirect extrusion

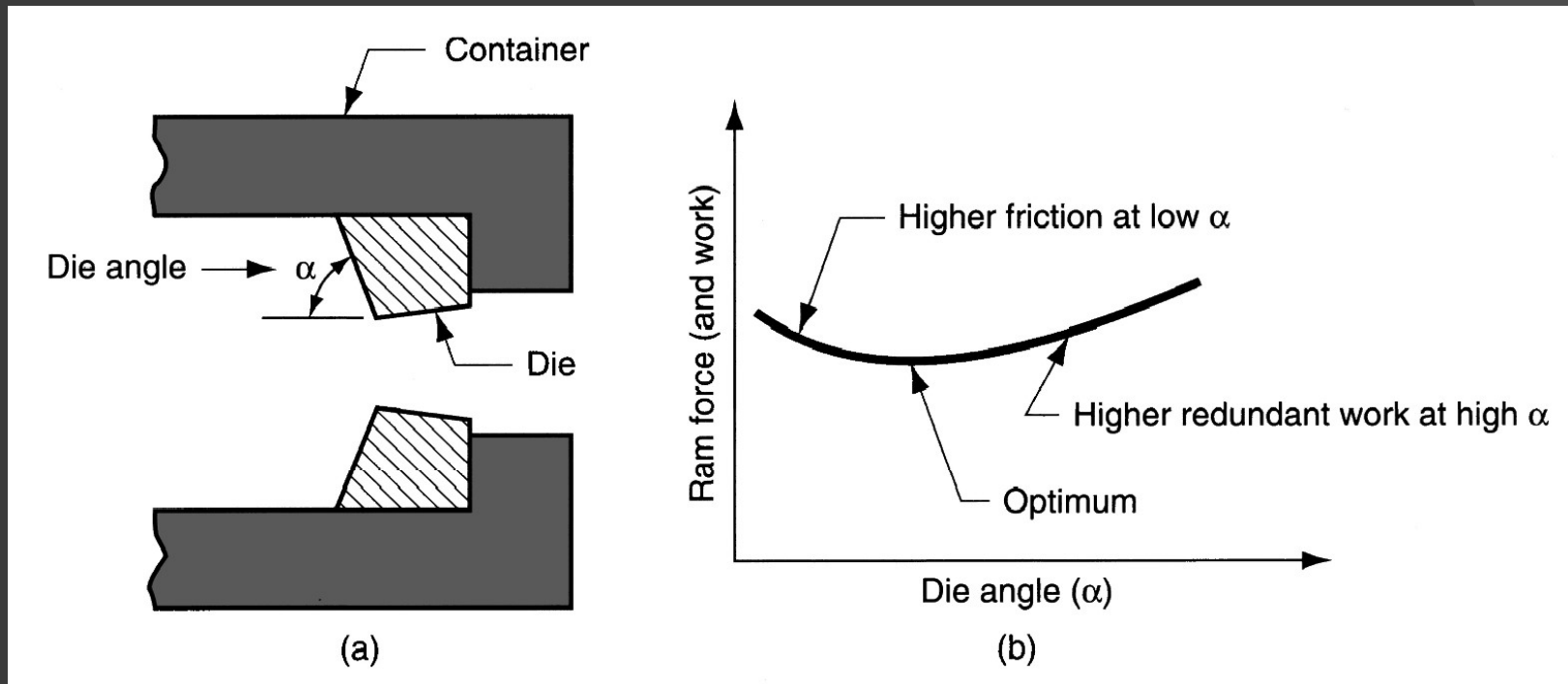


Figure -
(a) Definition of die angle in direct extrusion;
(b) effect of die angle on ram force

Comments on Die Angle

- Low die angle - surface area is large, leading to increased friction at die-billet interface
 - Higher friction results in larger ram force
- Large die angle - more turbulence in metal flow during reduction
 - Turbulence increases ram force required
- Optimum angle depends on work material, billet temperature, and lubrication

Comments on Orifice Shape of Extrusion Die

- Simplest cross section shape = circular die orifice
- Shape of die orifice affects ram pressure
- As cross-section becomes more complex, higher pressure and greater force are required

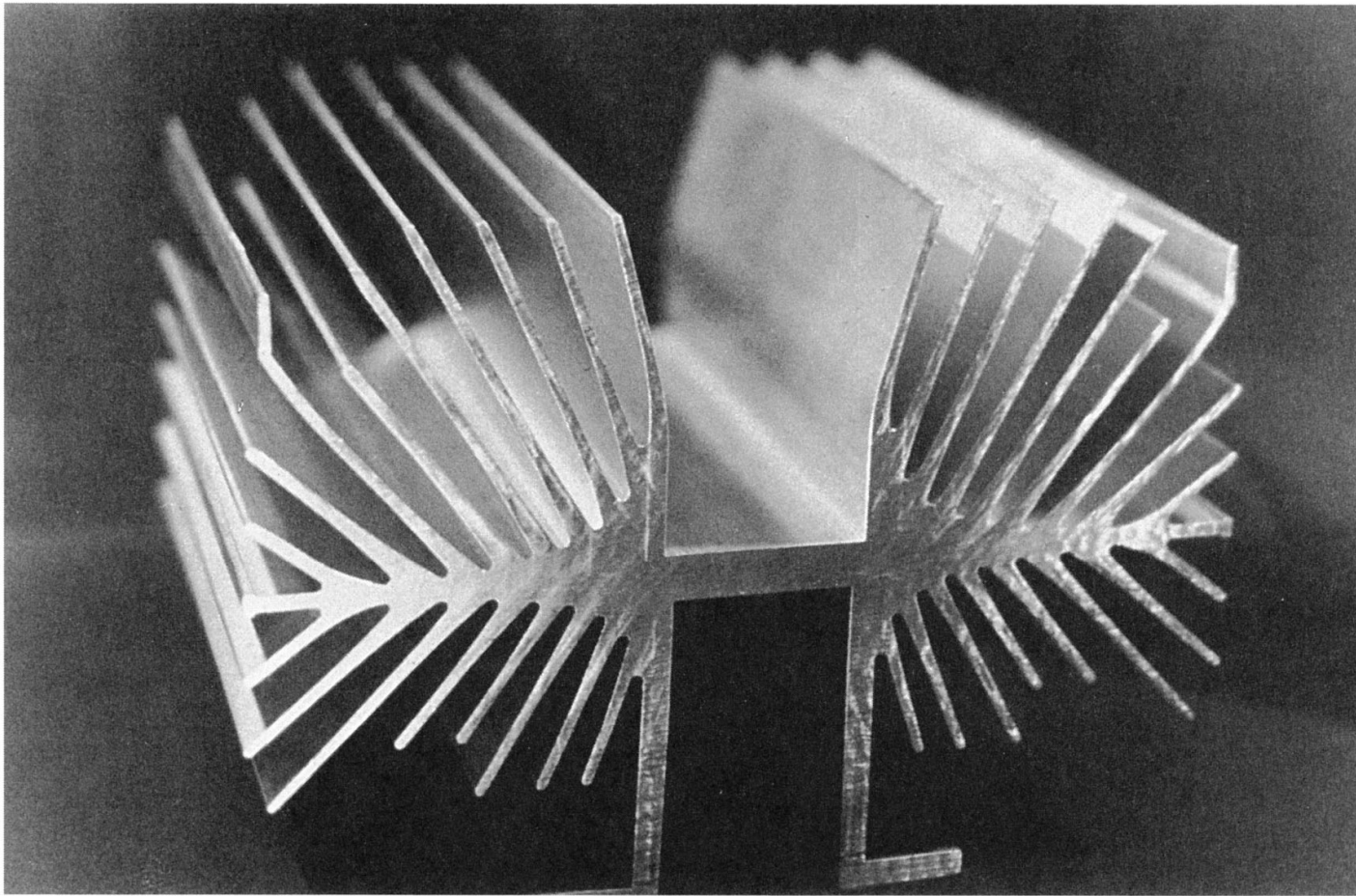


Figure - A complex extruded cross-section for a heat sink
(photo courtesy of Aluminum Company of America)

Extrusion Presses

- Either horizontal or vertical
 - Horizontal more common
- Extrusion presses - usually hydraulically driven, which is especially suited to semi-continuous direct extrusion of long sections
- Mechanical drives - often used for cold extrusion of individual parts

Wire and Bar Drawing

Cross-section of a bar, rod, or wire is reduced by pulling it through a die opening

- Similar to extrusion except work is *pulled* through die in drawing (it is *pushed* through in extrusion)
- Although drawing applies tensile stress, compression also plays a significant role since metal is squeezed as it passes through die opening

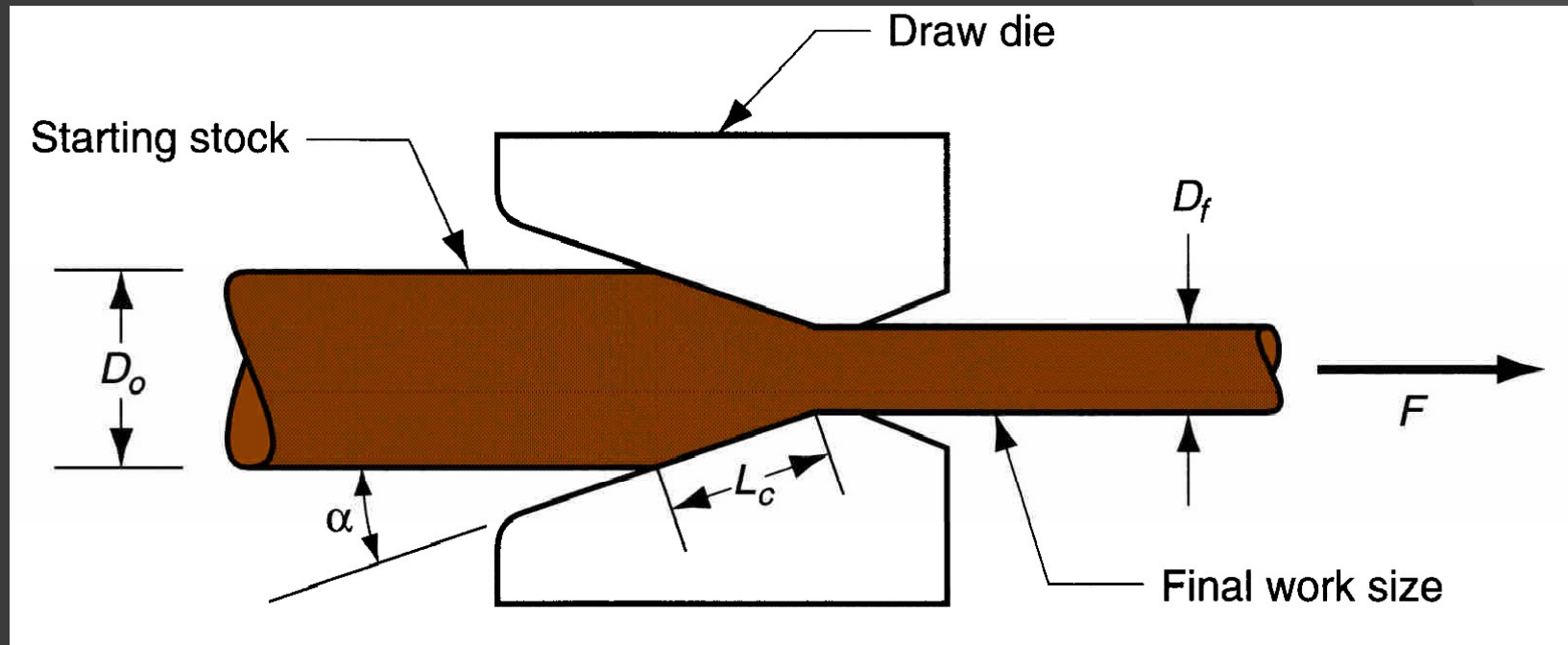


Figure - Drawing of bar, rod, or wire

Area Reduction in Drawing

Change in size of work is usually given by area reduction:

$$r = \frac{A_o - A_f}{A_o}$$

where r = area reduction in drawing; A_o = original area of work; and A_f = final work

Wire Drawing vs. Bar Drawing

- Difference between bar drawing and wire drawing is stock size
 - *Bar drawing* - large diameter bar and rod stock
 - *Wire drawing* - small diameter stock - wire sizes down to 0.03 mm (0.001 in.) are possible
- Although the mechanics are the same, the methods, equipment, and even terminology are different

Drawing Practice and Products

- Drawing practice:
 - Usually performed as cold working
 - Most frequently used for round cross-sections
- Products:
 - *Wire*: electrical wire; wire stock for fences, coat hangers, and shopping carts
 - *Rod stock* for nails, screws, rivets, and springs
 - *Bar stock*: metal bars for machining, forging, and other processes

Bar Drawing

- Accomplished as a *single-draft* operation - the stock is pulled through one die opening
- Beginning stock has large diameter and is a straight cylinder
- This necessitates a batch type operation

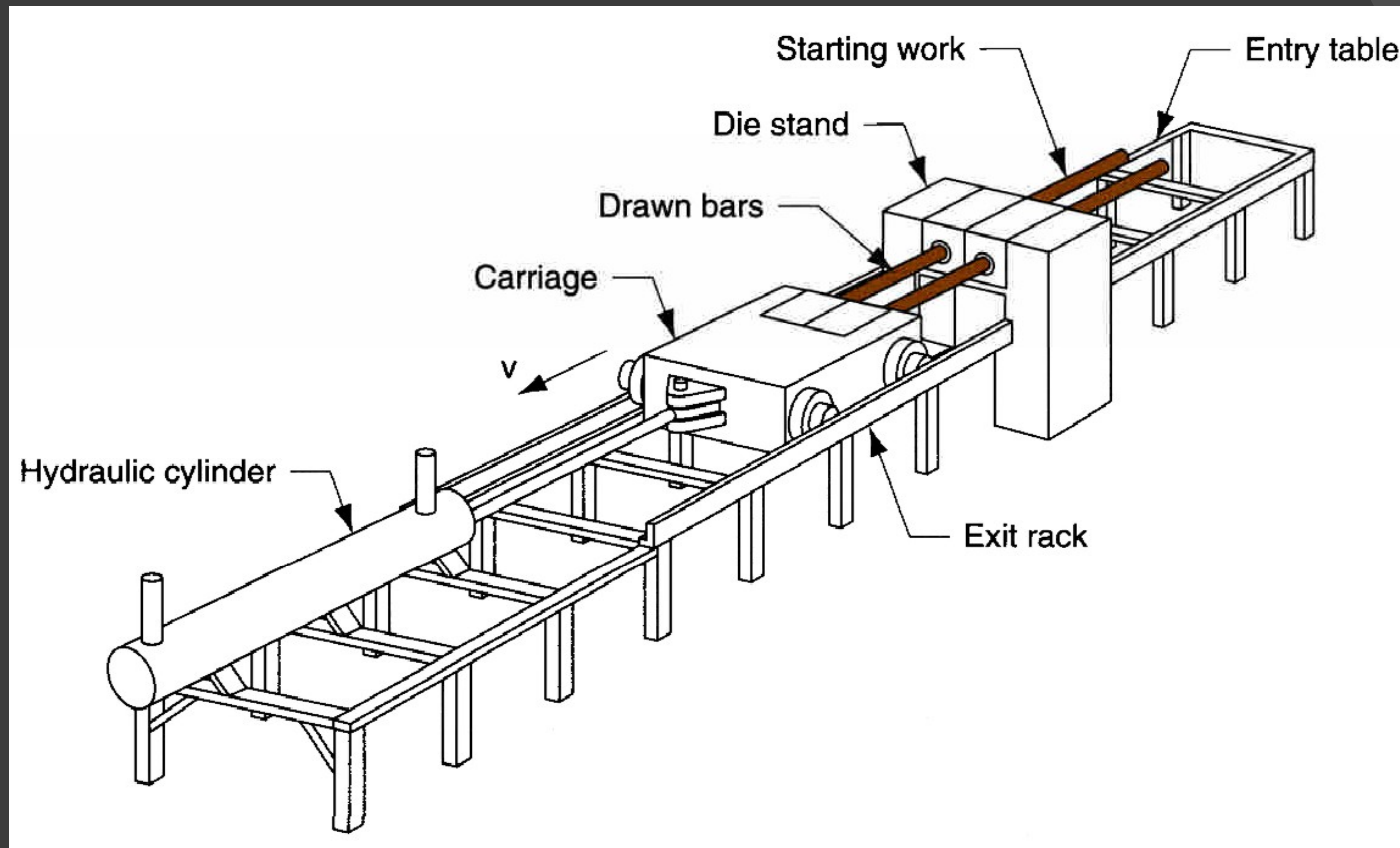
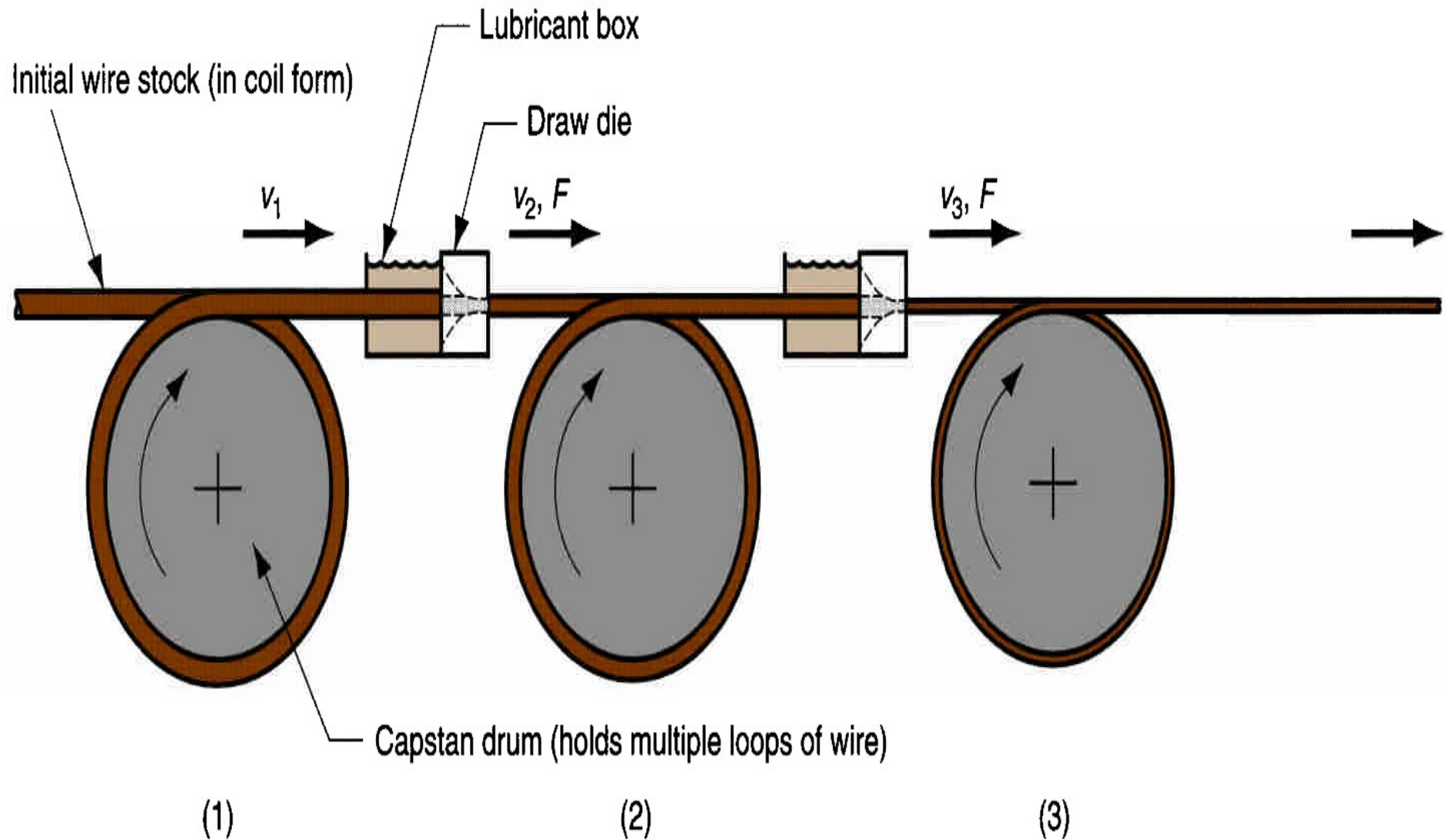


Figure - Hydraulically operated draw bench for drawing metal bars

Wire Drawing

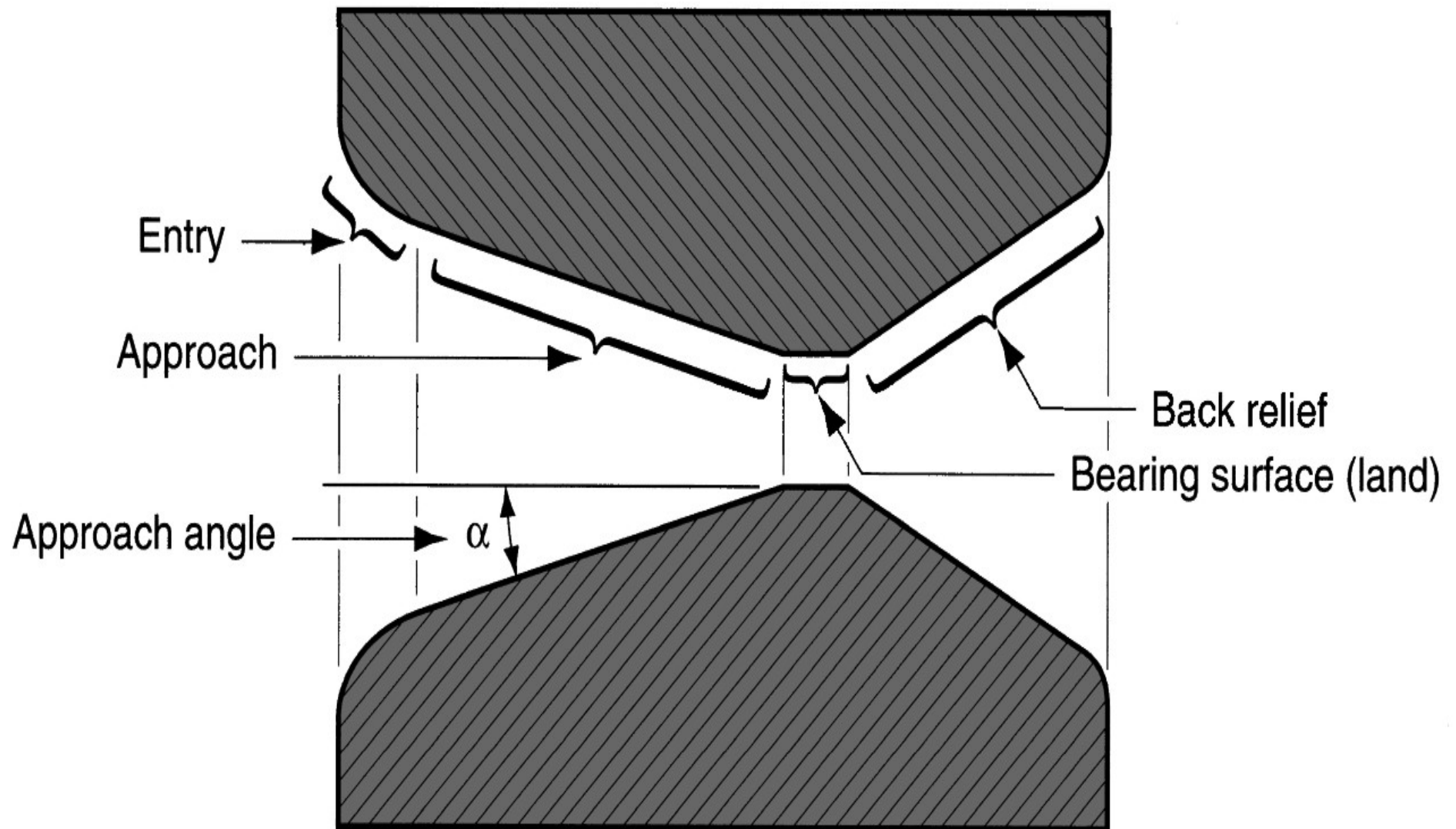
- Continuous drawing machines consisting of multiple draw dies (typically 4 to 12) separated by accumulating drums
 - Each drum (*capstan*) provides proper force to draw wire stock through upstream die
 - Each die provides a small reduction, so desired total reduction is achieved by the series
 - Annealing sometimes required between dies



Continuous drawing of wire

Features of a Draw Die

- *Entry* region - funnels lubricant into the die to prevent scoring of work and die
- *Approach* - cone-shaped region where drawing occurs
- *Bearing surface* - determines final stock size
- *Back relief* - exit zone - provided with a back relief angle (half-angle) of about 30°
- Die materials: tool steels or cemented carbides



Draw die for drawing of round rod or wire

Preparation of Work for Wire or Bar Drawing

- *Annealing* – to increase ductility of stock
- *Cleaning* - to prevent damage to work surface and draw die
- *Pointing* – to reduce diameter of starting end to allow insertion through draw die